

CHAPTER OUTLINE

PERITONEAL MEMBRANE ANATOMY AND STRUCTURE, 2094
 PERITONEAL TRANSPORT PHYSIOLOGY, 2095
 EVALUATION OF PERITONEAL SOLUTE AND WATER TRANSFER RATE, 2096
 THE PERITONEAL CATHETER AND ACCESS, 2097
 CATHETER-RELATED COMPLICATIONS, 2099
 PERITONEAL DIALYSIS SOLUTIONS, 2099
 PERITONEAL DIALYSIS MODALITIES, 2102
 DIALYSIS ADEQUACY, 2102
 CLEARANCE TARGETS AND CLINICAL OUTCOMES, 2105

NUTRITION AND PERITONEAL DIALYSIS, 2106
 INFLAMMATION AND PERITONEAL DIALYSIS, 2108
 RESIDUAL KIDNEY FUNCTION AND PERITONEAL DIALYSIS, 2108
 CARDIOVASCULAR DISEASE IN PERITONEAL DIALYSIS, 2109
 PERITONITIS, 2110
 NONINFECTIOUS COMPLICATIONS OF PERITONEAL DIALYSIS, 2114
 PATIENT OUTCOMES WITH PERITONEAL DIALYSIS, 2115

PERITONEAL MEMBRANE ANATOMY AND STRUCTURE

The peritoneum is a serous, semipermeable membrane constituted of a thin layer of connective tissue covered by a mesothelial cell monolayer, which covers most of the abdominal wall and intraabdominal organs. This mesothelium derives from mesenchymal cells which form a basement membrane, develop tight junctions, and desmosomes.¹ At the peritoneal cavity side, the mesothelial cells have abundant microvilli, which have anionic fixed charges and play a role in the transmembrane transfer of small charged molecules as well as plasma proteins.¹⁻⁴ The surface area of the human adult peritoneum is variable and ranges from 1.6 to 2.0 m², yet microvilli increase total peritoneal surface up to 40 m².⁵ Loss of microvilli is a common morphological change of the peritoneal membrane in patients receiving peritoneal dialysis (PD). When the peritoneal membrane is injured and mesothelial cells are undergoing apoptosis, surface anionic charges are reduced.⁶

The peritoneal microvessels and mesothelium are thought to function by either a two- or three-pore size model of capillary permeability.^{7,8} The three pore model pore sizes are: more than 150 Å for large pores, up to 40 to 45 Å for small pores, and 2 to 5 Å for ultra-small pores. The main pathway for exchanges across the microvascular wall is the junctions between capillary endothelial cells.

Mesothelial cells of the peritoneum have glucose transporters which play an active role in solute transport.⁹ In addition,

mesothelial cells express aquaporin channels, which correspond physiologically to the “ultra-small pores.” These channels may be modulated by diverse stimuli, including stimuli that are osmotic and nonosmotic.⁸ Desmosomes have also been observed near the cellular luminal front and gap junctions.¹⁰

The presence of mesothelial cells with stomata of 4 to 12 μm that communicate between the abdominal cavity and the submesothelial diaphragmatic lymphatics have been demonstrated.¹¹ Movement through this pathway of very large particles, as large as red blood cells, malignant cells, bacteria, and others have been demonstrated.^{12,13} Actin-like filaments of stomatal mesothelial cells, their channels, and lymphatic endothelial cells induce cell contraction and allow the passage of macromolecules and cells.

Under the mesothelial cell layer of the peritoneum there is a mono-layered structure denominated by the basement membrane, which has anionic charges along both the lamina rara externa and lamina rara interna.^{1,3} After long-term treatment with PD this basement membrane may duplicate and this phenomenon may be induced by cell death and by local exposure to high glucose concentration.¹⁴

The peritoneal interstitium (variable in thickness from 1 to 30 μm) is composed mainly of fibroblasts, collagen fibers, and by an amorphous proteinaceous substance (glycosaminoglycans, mainly hyaluronic acid), which displays anionic charges. Additionally, macrophages and mast cells are often, and monocytes rarely, present in this structure.^{3,15} Solute movement across the interstitial tissue is modified by diverse factors including thickness of the interstitium in that specific

site as well as molecular weight, shape, and electric charge of the molecule.¹⁶ To modulate the flow of water from plasma to lymph and to prevent interstitial edema, the local interstitial pressure is usually low and, at times, negative (0 to −4 mm Hg).¹⁷ Transfer of small solutes along the interstitium that is in general diffusive yet convective transport also contributes in the parietal peritoneum.¹⁸ Intraabdominal pressure plays an important role in the movement of fluid from the cavity to the interstitium. During PD, a positive intraabdominal pressure from 4 to 10 cm of H₂O drives fluid and solutes from the cavity to the interstitium and fluid loss is directly proportional to intraabdominal pressure.¹⁹

A thin basement membrane of the endothelial capillaries separates them from the connective tissue of the interstitium. Capillaries with fenestrae are present in the peritoneum and these structures as well as intercellular junctions play an important role in their permeability.²⁰ Tight junctions (zonula occludens) link the endothelial cells in a monolayer structure.²¹ In addition, the arteriolar endothelium typically exhibits gap junctions. There is significant controversy on which is the main pathway for water as well as small and large solutes across the peritoneal membrane. While some consider the intercellular cleft as the main pathway, others believe that tight junctions offer a paracellular barrier that regulates movement of water, solutes, and even immune cells between the interstitial space and the microvascular compartment.^{22,23}

When the presence of aquaporin-1 channels was demonstrated, it was clear that water transport occurs through several pathways. In addition to its transport via the paracellular pathway through intercellular junctions at the endothelial level, transcellular water transport was confirmed and a significant proportion of water transport flows through these ultra-small transcellular pores.^{24,25}

PERITONEAL TRANSPORT PHYSIOLOGY

In PD, solute transport occurs through both diffusion and convection. Diffusion takes place given the presence of a concentration gradient across a semipermeable membrane. The first law of Fick (the transfer rate of a solute is determined by the diffusive permeability of the membrane to that solute, the surface area available for transport, and the concentration) governs diffusion across the peritoneal membrane. A second mechanism involved in solute transport is convection, which takes place during ultrafiltration. This type of transport is determined by the mean concentration of the solute, water flux, and the specific solute reflection coefficient of the membrane.²⁶ A factor that influences transport across the peritoneal membrane is the effective surface area available for this process to happen and is strongly determined by the number of capillaries and the proportion of the peritoneal membrane in contact with the dialysate. In addition, the peritoneal membrane has a defined intrinsic permeability, which determines the features of solutes to be transported.²⁷

The peritoneum poses several barriers to solute transport.²⁸ Peritoneal capillaries are the main barrier. A two-pore theory proposes the presence of abundant small pores of 40 to 50 Å and few larger pores of up to 150 Å.²⁹ With the confirmation of the presence of transcellular (endothelial cells) ultra-small pores (aquaporin-1 channels, 3 to 5 Å) it has been

shown that around 50% of the transcapillary ultrafiltration occurs through them.^{25,30} This led investigators to propose a three-pore model, which has shown that around 50% of the transcapillary ultrafiltration occurs through aquaporin-1 channels and the remaining mostly through small pores.^{31,32} The interstitium constitutes a significant barrier to water and solute transport across the peritoneal membrane.^{33–35} In order to describe the kinetics of water and solute movement during PD, several distributed models have been proposed.^{34,36–38}

Solute transport along time is highly dependent on the ultrastructural characteristics of peritoneal capillaries. Transport of low and medium molecular weight solutes depends mostly on their size and the surface area available for transport and to a lesser degree to changes of permeability of the, and peritoneum.³⁹ Transport of larger molecules is size-selective therefore depends on several factors including effective membrane surface area, permeability of the membrane itself, and on the molecular size of the solute.^{31,40} While in animal models the anionic negative charge at the peritoneal barrier may restrict macromolecular clearance, this phenomenon has not been demonstrated in humans.^{40,41} In order to calculate the mass transfer area coefficient (MTAC), which is the theoretical instantaneous maximal clearance at time 0 without ultrafiltration, models of variable complexity have been proposed.^{38,42} For PD clinical practice, simple procedures that correlate appropriately with MTAC have been developed. For example, a 24-hour clearance or a 4-hour dialysate/plasma (D/P) ratio of low molecular weight solutes is clinically used to evaluate dialysis efficiency.⁴³ As described above, diffusive transport accounts for the majority of low molecular weight removal (e.g., potassium, urea, and creatinine) and convection represents a small proportion of solute transport.^{44,45}

Peritoneal water transport is mostly driven through an osmotic gradient generated by additives of the PD solution, generally glucose or icodextrin. The normal peritoneal membrane also presents a small and continuous amount of transcapillary ultrafiltration to the peritoneal cavity through differences in hydrostatic and colloid osmotic pressures and via small and ultra-small pores. Coupled to this transcapillary ultrafiltration there is reabsorption of water from the peritoneal cavity through transcapillary back-filtration and peritoneal lymphatics.⁴⁶ Given these physiologic pathways of water transport, the net ultrafiltration achieved in a PD exchange is the balance between the transcapillary ultrafiltration into the cavity and back absorption through capillaries and lymphatics.

As described above, the osmotic reflection coefficient of a given dialysate solution determines its effectiveness to induce ultrafiltration. Large molecules such as proteins are not permeable and have a reflection coefficient of 1, while very small solutes have a reflection coefficient close to 0. The reflection coefficient may be different according to the size of the pores; therefore glucose through “small pores” is very low and close to 0, while it is very high (value of 1 and impermeable) through aquaporin-1 ultra-small pores.^{47–49}

Water transport during a standard glucose-based dialysate can be described as follows^{44,46}: When dialysate is infused (time 0), glucose concentration in the dialysate is highest and therefore, crystalloid osmotic pressure and ultrafiltration rate are also highest. Glucose is absorbed from the dialysate over time (almost two thirds of glucose content of a solution over a 4-hour period), and the crystalloid pressure and

ultrafiltration diminish.⁵⁰ Ultrafiltration volume accumulates within the peritoneal cavity and will peak at the time there is osmotic equilibrium between serum and dialysate. This process will take place along with a progressive reduction of transcapillary ultrafiltration rate, to the point in which it is equal to the rate of lymphatic and back absorption of the capillaries. From then on, and if the lymphatic and back absorption rates exceed the transcapillary ultrafiltration rate, intraperitoneal volume will diminish. Patients will display individualized transport characteristics that can be explored with a clinical peritoneal transport test (see the following section).⁴⁴

EVALUATION OF PERITONEAL SOLUTE AND WATER TRANSFER RATE

In patients with end-stage renal disease, maintenance dialysis allows for the removal of solutes and water that would otherwise be excreted by the kidneys. The efficiency of solute and water removal depends, in part, on the characteristics of the dialysis membrane. In patients treated with PD, the naturally occurring peritoneal barrier serves as the dialysis membrane. It has long been recognized that there is a large inter-individual variability in the efficiency of transfer of solutes and water across the peritoneal barrier and in a significant proportion of individuals the efficiency and other transport characteristics change over time with exposure to PD solutions.^{44,51} Hence, it is imperative to characterize the peritoneal solute and water transfer rate to individualize the dialysis prescriptions. This testing is often performed at the time of initiation of dialysis or soon thereafter and repeated if clinically indicated.

Several approaches have been developed to characterize the rate of solute and water transfer in patients undergoing PD. These include the Peritoneal Equilibration Test (PET), Standard Permeability Analysis (SPA), Peritoneal Dialysis Capacity (PDC), and Dialysis Adequacy and Transport Test (DAT).^{44,50,52,53} Each of these tests allow for a standardized assessment of solute transfer and ultrafiltration capacity of the peritoneum, information that is critical to fashion an effective PD prescription (Table 64.1). Often, the solute transfer function and ultrafiltration capacity as assessed by each of these tests is not the same as a precise measure of diffusion and convection across the peritoneum. Common

to these approaches is that each step of the procedure is performed in a standardized way. This standardization is critical in ensuring the reproducibility of the results, essential for its clinical application.

The PET is the most widely used measure of assessment of peritoneal solute and water transfer rates because of overall ease of use and interpretation in day-to-day clinical practice. The key elements of the process that were standardized when the test was first described included having a long preceding overnight exchange (8-12 hours), method to completely drain the overnight exchange prior to start of test (sitting position, over 20 minutes), volume and concentration of dialysate instilled (2L of 2.5% dextrose), rate of infusion (400 mL/2 minutes; total infusion time, 10 minutes), total dwell time (240 minutes), timing of collection of dialysate (0, 30, 60, 120, 180, and 240 minutes), time for venipuncture (120 minutes), and the way the results of the test are expressed (4-hour dialysate to 2-hour plasma creatinine, “D/P” creatinine, or similarly for other solutes; 4-hour to 0-hour dialysate glucose, D/D0 glucose; and ultrafiltration volume).⁴⁴ Since the first description of the test, the standardized test has been modified for collection of dialysate at 0, 120, and 240 minutes (modified PET). A large number of other variations to this initial description of the modified PET have been described and include variation in length of preceding exchange, dwell time (60 minutes), tonicity of dialysate (4.25% dextrose), or the use of biocompatible PD solutions.⁵⁴⁻⁶⁰ Still, the overwhelming majority of patients starting treatment undergo the modified PET with 2 L 2.5% dextrose.

As noted above, there is a large inter-individual variability in the rate of transfer of solute and water across the peritoneum, requiring assessment of peritoneal function in each patient. Even though the PET can describe the rate of transport of virtually any solute, the 4-hour D/P creatinine is the parameter most widely used both for clinical and research purposes. Large population-based studies suggest that, on average, the dialysate concentration of creatinine at four hours is about two thirds that in the blood (or 4-hour D/P creatinine, 0.67).⁴⁴ It is conventional to assign patients a “transport” type based on the results of the PET as low, low-average, high-average, or high (or slow, slow-average, fast-average, and fast) transporters.⁴⁴ Individuals assigned as having an average transport are the ones that are within one standard deviation of the mean and the ones with slow or fast transport are more than one standard deviation below or above the mean, respectively.

Variability in the peritoneal solute transfer rate likely reflects, for the most part, the density of peritoneal capillaries per unit surface area. Simply, the 4-h D/P creatinine could be considered as a measure of the effective peritoneal surface area. The reason for the large inter-individual variability in peritoneal solute transfer rate is not clear. Demographic and clinical variables (such as age, sex, race, diabetes, cardiovascular disease) explain only a very small fraction of this variability.⁶¹ Preliminary studies indicate that some of this variability is, in part, genetically determined.⁶²⁻⁶⁸ There is also compelling evidence for intra-peritoneal inflammation as measured by dialysate IL-6 levels as an important determinant of peritoneal solute transfer rate; this may, in part, be also genetically determined.^{62,69,70} Large population-based genome wide association studies are presently underway that could help clarify the genetic underpinnings in the variability

Table 64.1 Parameters Obtained From Standardized Tests to Evaluate Peritoneal Membrane Function

Test	Solute	Fluid Removal
Peritoneal equilibration test	D/P creatinine D/D0 glucose	Drain volume
Standard permeability analysis	Mass transfer area coefficient, creatine	Drain volume D/P sodium
Peritoneal dialysis capacity	Area parameter	Ultrafiltration coefficient
Dialysis adequacy and transport test	24-hour D/P creatinine	24-hour drain volume

of peritoneal solute transfer rate; this, in turn, could help us better understand peritoneal biology and pathobiology.

The information obtained from a PET could be used as a guide to optimize PD prescription; several validated software programs based upon urea kinetic modeling have been developed that use data from the PET to optimize prescriptions.^{71,72} The slower the peritoneal solute transfer rate, the greater is the challenge with solute removal particularly in large, muscular anuric individuals; inadequate ultrafiltration is rarely, if ever, a concern. These individuals need longer dwell times to achieve adequate solute removal, which can be done both with continuous ambulatory or automated PD. In contrast, the faster the solute transport, the greater is the challenge with fluid removal, particularly with loss of residual kidney function; achieving an adequate dose of dialysis is rarely, if ever, a concern. These individuals generally need shorter dwell times with a cyclor at night and extreme care in limiting long dwell times with dextrose-based solutions. The long dwell times can be optimized with either complete or partially “dry” days in individuals with significant residual kidney function (RKF), or with the addition of daytime exchange or icodextrin in individuals without RKF.

In addition, a large number of studies have examined the association of peritoneal solute transfer rate with meaningful patient outcomes. The preponderance of evidence seems to suggest that patients with higher peritoneal solute transfer rates have a higher risk for death.^{73–78} Evidence linking peritoneal solute transfer rate with other meaningful outcomes, such as risk for transfer to hemodialysis or protein-energy wasting, is inconsistent. Several hypotheses have been put forth to explain the higher risk for death in high/fast transporters and include a higher prevalence of protein-energy wasting from greater protein losses and/or suppression of appetite from greater absorption of glucose, systemic inflammation, or comorbidity. However, there is an emerging consensus that the higher risk for death in high/fast transporters is from inadequate ultrafiltration with conventional PD prescriptions, particularly with continuous ambulatory PD.⁷⁹ Even though the risk is mitigated with use of automated PD with shorter nighttime dwells, the risk is not abrogated.⁷⁸ Thus, it is imperative that care be observed when designing prescriptions that optimize fluid removal in high/fast transporters.

THE PERITONEAL CATHETER AND ACCESS

Successful PD therapy depends upon permanent and safe access to the peritoneal cavity. A well-functioning catheter provides obstruction-free access to the peritoneum. Mechanical catheter problems cause transfer to HD in nearly 20% of prevalent PD patients; peritoneal infections, which are often catheter-related, are responsible for another 30% to 50% of PD technique failures.^{80,81} Therefore, proper catheter selection and placement is imperative to the success of PD.

Globally, the most widely used catheter is the standard Tenckhoff catheter followed by the swan-neck catheter.⁸² Catheters are made either of polyurethane or silicone rubber. Polyurethane, a stronger material than silicone rubber, allows creation of a thinner catheter wall resulting in a larger catheter lumen, which in turn allows faster flow rates (3.1 mm polyurethane vs. 2.6 mm internal diameter traditional silicone).

More recently, some manufacturers have begun offering larger diameter silicone catheters in some countries (e.g., Flex-neck catheter, Cardiomed). Polyurethane catheters are most often found outside of the US and can degrade with the routine use of ointments and povidone-iodine at the exit site, whereas silicone catheters have been reported to degrade with exposure to povidone-iodine but not ointments such as mupirocin.⁸³

PD catheters have essentially three segments: the intraperitoneal, tunneled, and extraperitoneal segment. The intraperitoneal segment is the portion of the catheter that resides inside the peritoneal cavity; side holes along the length of the intraperitoneal portion deliver and drain dialysate. The extraperitoneal segment passes through a subcutaneous tunnel within the abdominal wall (intramural), exits through the skin, and has an external (outside the body) segment. The catheter may contain one or two polyester cuffs, typically 1 cm long, referred to as the internal (pre-peritoneal) and external (subcutaneous) cuffs. Fig. 64.1 shows different intraperitoneal and extraperitoneal designs of currently available peritoneal catheters.

The straight double-cuff Tenckhoff catheter is available in variable lengths but is most often used in either the 42 cm or 47 cm length, with an intraperitoneal segment length of 15 cm in the former and 20 cm in the latter; both have an intramural segment about 5 to 7 cm long, and an external segment about 20 cm long.^{84,85} The intraperitoneal segment has multiple 0.5-mm side openings along its course, the length of which measures 10–15 cm depending on the total catheter

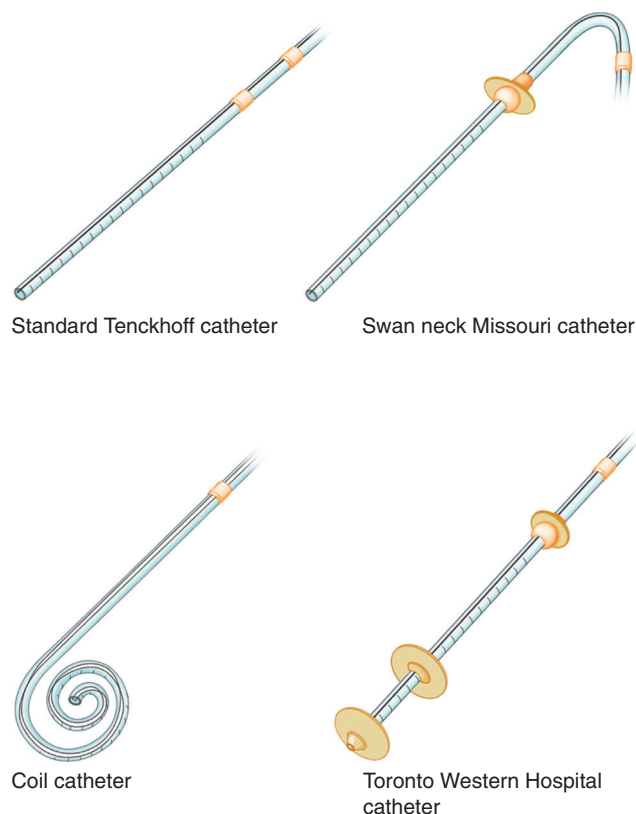


Fig. 64.1 Intraperitoneal and extraperitoneal designs of currently available peritoneal catheters.

length. The coiled Tenckhoff catheter is also available in variable lengths between 57 cm and 72 cm with a coiled, perforated intraperitoneal end that is 19.5 cm long for all coiled catheters. It is important to select the proper catheter length for each patient individually to avoid malposition of the catheter in the peritoneum; too-short catheters are at risk for incomplete drainage and entrapment from hanging omentum while too-long catheters can lead to inflow pain most often in the rectal or perineal region. Most Tenckhoff catheters have a barium-impregnated radiopaque stripe throughout the catheter length to assist in radiologic visualization.

Several different configurations of peritoneal catheters exist beyond the straight and coiled-tip Tenckhoff catheter. The swan-neck catheter, a modified Tenckhoff catheter, features a preformed 180-degree bend between the two cuffs.^{82,86} These catheters can be placed in an arcuate tunnel such that both external and internal segments of the tunnel can easily be directed downward. The preformed angle between cuffs eliminates the risk of the catheter straightening over time due to rubber “shape memory,” thus potentially reducing the risk for internal catheter tip migration and external cuff extrusion.⁸⁷ Swan-neck catheters are available with three different intraperitoneal configurations: straight, coiled, and straight with two intraperitoneal silicone discs (also called the Toronto-Western or Oreopoulos-Zellerman catheter). The coiled and silicon-disk configurations were each developed to help reduce omental wrapping and to maintain catheter position in the pelvis. The Toronto-Western and Missouri catheters also differ from the Tenckhoff catheter because they replace the internal cuff with a felt disc-silicone bead combination; the disk is sutured to the rectus muscle just outside the peritoneum and serves as an anchor for the catheter while the bead is placed just inside the peritoneum and serves as a physical barrier to prevent peritoneal fluid leakage. All of the aforementioned catheters are available with the traditional subcutaneous cuff that is optimally placed 2 cm proximal to the catheter exit site.

Presternal catheters, another modification of catheter design, enter the peritoneum in the traditional location and are then tunneled subcutaneously and ipsilaterally up to the chest wall, where they finally exit the skin. The unique design of the presternal catheter offers several benefits over standard abdominal catheters in certain settings. Presternal exit sites are easily visible even in the obese patient who has a large pannus, or other visual obstruction to the lower abdominal skin. Patients with limited flexibility who cannot easily bend to see a traditional exit site also benefit from the easy visibility of a presternal exit site. The presternal catheter allows patients with abdominal ostomies and pediatric patients in diapers to utilize PD without risk of exit-site cross-contamination. Presternal catheters potentially reduce the risk of catheter trauma (a risk factor for exit-site infection) because the chest is a rather rigid structure with minimal wall motion. A long catheter tunnel, combined with three cuffs, may reduce pericatheter bacterial contamination of the peritoneal cavity and hence reduce the incidence of peritonitis.⁸⁸

The presternal catheter is composed of two silicone rubber tubes, cut to an appropriate length and connected end to end at the time of implantation. The intraperitoneal tube is available in the dual cuff or the Missouri catheter configuration while the upper tube is a variation on the swan-neck

catheter. A titanium connector connects the two components at the time of implantation. The catheter should be tunneled ipsilaterally to the peritoneal insertion point so that implantation directly over the sternum is avoided, therefore preventing catheter damage during any cardiac surgery that would necessitate sternotomy. Outcomes with the presternal catheter have been favorable in experienced hands.⁸⁸

The Moncrief-Popovich (embedded) catheter is a modified swan-neck coiled catheter with a longer subcutaneous cuff (2.5 cm instead of 1 cm) that is implanted normally except that the external portion of the catheter is buried subcutaneously at the time of placement. This catheter is inserted months before it is needed, typically before the patient requires dialysis. The embedded catheter heals in a sterile environment thus there is low risk for early bacterial catheter colonization, in turn reducing the risk for exit site or peritoneal infection.^{89,90} When dialysis is needed, the external portion of the catheter is exteriorized and connected to an external catheter extender in a short outpatient procedure. Full inflow volumes can be used immediately without increased risk of catheter leak because the catheter has healed fully over the preceding months while embedded. A futility rate of approximately 10% has been reported with embedded catheters (due to transplantation, death, or treatment with hemodialysis instead of PD); nevertheless, overall embedded catheters that are needed tend to work well. Early catheter drain problems have been reported to occur in 5% to 29% of exteriorized embedded catheters, are most often due to fibrin, and can be readily corrected without catheter removal.^{91–94}

Rigid catheters for acute dialysis, rarely used in developed nations, are still used in some countries. Complications of rigid catheter insertion include minor bleeding, leakage of dialysis solution, extravasation of fluid into the abdominal wall, particularly in patients who have had a previous abdominal surgery or multiple catheter insertions, and inadequate drainage as a result of omental wrapping, loculation, or misplaced catheter in the upper abdomen. Loss of a part or the entire rigid catheter after manipulation of a poorly functioning rigid catheter has been reported. The incidence of peritonitis varies widely with rigid catheters; the rate may be dependent on the duration of dialysis and history of catheter manipulation among other factors.^{95,96}

For long-term use, standard non-rigid PD catheters can be inserted by nonsurgical or surgical methods. PD catheter placement can occur at the patient's bedside by using Seldinger technique (blind insertion); however this method incurs a nontrivial risk of bowel trauma. Peritoneoscopic-assisted catheter placement, often used by nephrologists, allows visualization of the peritoneum before catheter insertion, thus limiting the risk of bowel trauma, but does not allow simultaneous visualization while the catheter is physically inserted. A modified percutaneous approach with fluoroscopic and ultrasound guidance at the time of catheter placement allows both avoidance of bowel and confirmation of catheter placement.⁹⁷ An experienced (interventional) nephrologist, interventional radiologist, or surgeon should be the operator in such cases. Surgical catheter placement was historically performed using open surgical techniques (mini-laparotomy) but technological advances have led to increased utilization of laparoscopic catheter placement techniques that offer the advantage of direct peritoneal visualization. Basic laparoscopic-guided PD catheter placement allows proper

catheter placement in the lower pelvis. Surgical placement via open mini-laparotomy has been increasingly replaced by surgical laparoscopic catheter placement because the latter allows visualization and correction of hernia, adhesions, and low-hanging omentum at the time of catheter insertion; additionally, laparoscopy ensures correct catheter tip placement in the pelvis.^{98,99} Advanced laparoscopic techniques such as rectus sheath tunneling, selective prophylactic omentopexy, and selective prophylactic adhesiolysis provides additional benefits at the time of catheter placement and in experienced hands has resulted in a 2-year catheter survival rate of 99.5%.^{98,100,101}

CATHETER-RELATED COMPLICATIONS

The most common complications of PD catheters include exit site and tunnel infection, impaired flow, external cuff extrusion, dialysate leaks, and infusion pain. Catheter-related infections can result from improper catheter placement, poor wound healing at the exit site, external catheter trauma, interference of the normal healing process during and after catheter placement, and inadequate routine exit-site care. Exit-site infections have been shown to be a result of sinus tract bacterial colonization; therefore it is imperative for a new exit site to heal in a sterile environment for as long as possible. One way to achieve this is by keeping the original post-surgical dressing in place for 5 to 7 days after catheter placement, removing it only in the case of frank bleeding or drainage; trained PD personnel should be the first to uncover the original catheter dressing and they should always wear masks while inspecting the exit site. After the exit site is well healed, trauma to the external segment of the catheter predisposes to infection by allowing bacterial colonization of the sinus tract and exit site and impairing normal tissue regeneration.^{102–106} Certain cleansing agents can also impair normal skin cell turnover and should be avoided (e.g., iodine, bleach, alcohol).

Impaired dialysate flow can be seen either during drainage, inflow, or both. Impaired catheter drainage is often associated with constipation, when stool-filled bowel loops either obstruct catheter side holes or cause loculated fluid collections that are not readily accessible for drainage. A successful laxative regimen, over the course of one day, is often successful in restoring proper catheter drainage; abdominal radiographs can confirm constipation when the clinical picture is unclear. Abdominal radiographs are also useful to detect PD catheter tip migration, which is another cause of impaired dialysate drainage. Catheter repositioning can be attempted non-invasively using guidewire manipulation under fluoroscopy, and if unsuccessful, laparoscopic catheter repositioning is typically successful in improving catheter function when performed by an experienced surgeon. As described earlier, the swan-neck catheter configuration may reduce the risk of catheter migration over time.¹⁰⁷

Entrapment, or “capture,” of the catheter by the active omentum may cause outflow obstruction in the post implantation period. Omental “capture” as a late event is rare. From time to time in some patients, drainage slows as a result of catheter translocation, obstruction by omentum, or fibrin clot formation. Laxatives or addition of heparin, 500 U per liter of dialysis solution, or both may be successful in restoring good dialysate flow. In some patients, catheters have migrated

out of the true pelvis. If the catheter continues to function appropriately, it is not necessary to reposition it. If the catheter fails to function after simple maneuvers are implemented, more aggressive measures (e.g., laxatives, forced flushing) may be tried. When these measures fail, laparoscopic repositioning of the catheter tip back to the true pelvis and anchoring may be necessary. The Toronto Western catheter has two silicone discs in the intraperitoneal segment of the catheter that hinder the free movement of catheter tip out of the pelvis after placement.¹⁰²

The catheter tip, as it rests against the pelvic wall or intraabdominal organs, may cause localized pain from irritation.¹⁰⁵ The jet effect of rapidly flowing dialysis solution may also cause abdominal pain. In some rare instances, compartmentalization from adhesion formation around the catheter may cause severe abdominal pain.¹⁰⁸ Coiled catheters are designed to reduce infusion-related abdominal pain by avoiding a direct jet effect of solution against the bowel.

The extrusion of the external synthetic cuff can be prevented by creating the tunnel in a shape similar to the shape of the catheter and placing this cuff approximately 2 to 3 cm from the skin. In the absence of catheter infection, shaving off the extruded external cuff may help prolong the life of the catheter.¹⁰⁴

Insertion of the deep cuff into the center of the rectus muscle, as opposed to midline placement, has significantly reduced the incidence of early leakage of pericatheter dialysis solution.^{102,105} Pericatheter leaks are rare with catheters that have a bead and polyester flange at the deep cuff (Toronto Western Hospital catheter, Swan Neck Missouri catheter, Swan Neck Presternal Peritoneal catheter). In contrast to the early leaks, which are usually external, the late leaks infiltrate the abdominal wall through prior healed incisions. PD catheters may cause hemoperitoneum by causing minor tears of small vessels. On occasion, a peritoneal catheter erodes into the mesenteric vessels, leading to hemoperitoneum. In rare cases, a peritoneal catheter damages the internal organs, which leads to intraabdominal bleeding. Transvaginal leakage of peritoneal fluid is rare, but the possibility should be considered in an appropriate clinical setting.^{102–105}

PERITONEAL DIALYSIS SOLUTIONS

The dialysate used to perform PD can be considered to have three components: (1) an osmotic agent to induce ultrafiltration; (2) buffer to correct uremic metabolic acidosis; and (3) a combination of electrolytes to optimize diffusive removal of solutes. The various options for each of these components among the solutions that are commercially available are listed in Table 64.2. The most widely used formulation comprises dextrose as an osmotic agent (1.5%, 2.5%, or 4.25%), lactate as the buffer (usually 40 mEq/L), physiologic concentration of calcium (2.5 mEq/L), at a pH of 5.4. Even though several decades of use support the general efficacy and safety of conventional PD solutions for the long-term treatment of end-stage kidney disease (ESKD), concerns about the limitations or concerns about each of these components has driven the quest for better solutions.

Conventional PD solutions contain dextrose (or glucose monohydrate) as the **osmotic agent**; three different strengths of solution are commercially available as 1.5%, 2.5%, or 4.25%

Table 64.2 Components of Different Peritoneal Dialysis Solutions Available Commercially

	Osmotic Agent	Buffer	pH
Conventional	Dextrose	Lactate	5.2
Low-glucose degradation product	Dextrose	Lactate or bicarbonate	7.0–7.4
	Dextrose	Lactate and bicarbonate	7.4
Icodextrin	Icodextrin	Lactate	5.2
Amino acids	Amino acids	Lactate	6.4

dextrose (or 1.36%, 2.25%, and 3.86% glucose, respectively). The supraphysiologic concentration of glucose in PD solutions exerts a high osmotic force across the peritoneal barrier to generate brisk ultrafiltration.⁴⁶ Yet, there are two major limitations. First, glucose is absorbed across the peritoneum and this reduces the osmotic force driving ultrafiltration over the course of an intraperitoneal dwell. Hence, ultrafiltration volume is often inadequate with glucose-based PD solutions with longer dwell times such as the overnight dwell in patients undergoing continuous ambulatory PD or the day dwell in patients treated with automated PD.¹⁰⁹ Second, a wide range of glucose degradation products form during heat sterilization of glucose-based PD solutions.^{110,111} Many of these degradation products are either directly cytotoxic or accelerate the formation of advanced glycosylation end-products and are thought to underlie the structural and functional changes seen in the peritoneum with long-term PD.¹¹² These limitations have led to development of solutions that contain either an osmotic agent better suited for longer dwell times (such as icodextrin), or glucose-based solutions with undetectable concentrations of glucose degradation products (biocompatible PD solutions). A 1% amino acid solution can also be used as an osmotic agent – alone, or in combination with 1.5% dextrose – and may be a useful adjunct in patients with protein-energy wasting (see below).

Lactate is the most commonly used **buffer** in PD solutions. During PD, bicarbonate in the blood moves into the dialysate, and lactate is absorbed systemically, each along its concentration gradient. Lactate is metabolized in the liver into bicarbonate and leads to correction of uremic metabolic acidosis. There are few, if any, adverse consequences from the use of lactate as a buffer. However, with the recognition that lactate is not a physiologic buffer, bicarbonate-based solutions have been developed. Such solutions have two compartments that separate calcium and magnesium from bicarbonate; this precludes precipitation of bicarbonate salts during heat sterilization of PD solutions. The two compartments are separated by a thin membrane that is disrupted prior to infusion of dialysate into the peritoneal cavity. The resultant solution has a physiologic pH, unlike the solution in single chamber bags such as dextrose- or icodextrin-based solutions.

Conventional PD solutions also have a combination of other **electrolytes** such as sodium, calcium, magnesium, and chloride. The concentration of sodium in PD solutions (132 mEq/L) is lower than in the dialysate used for HD (138–142 mEq/L); this allows for a somewhat greater diffusive

removal of sodium. In order to further enhance sodium removal, PD solutions with significantly lower concentrations of sodium (102–125 mEq/L) have been tested. Such low sodium PD solutions may require higher concentration of glucose to ensure an adequate osmotic force across the peritoneal barrier and result in greater sodium removal with resultant better control of blood pressure than conventional PD solutions.^{113,114} However, a recent multicenter trial has raised concern about lower residual kidney function in patients randomized to low-sodium dialysate (125 mEq/L).¹¹⁴ As such, low-sodium dialysate remains experimental thus far and is not commercially available.

Older formulations of PD solutions contained a higher concentration of calcium (3.5 mEq/L). Supraphysiologic calcium concentrations can help to control hyperparathyroidism in some patients, but can place others at risk for hypercalcemia and oversuppression of parathyroid hormone, especially if treated with calcium-based phosphate binders and/or calcitriol of active vitamin D analogs. In order to mitigate this risk, most PD solutions used in contemporary clinical practice contain physiologic concentrations of calcium (2.5 mEq/L).^{115,116}

The following paragraphs provide an overview of three different PD solutions available commercially in different parts of the world (icodextrin, low-glucose degradation product solution, and bicarbonate-based solution) and the role for glucose-sparing regimens; the clinical experience with amino acid solutions is discussed along with other considerations for management of protein energy wasting.

ICODEXTRIN PD SOLUTION

Icodextrin is a polymer in which molecules of glucose are linked by alpha (1–4) bonds.¹¹⁷ It is derived from hydrolysis of corn starch, each chain has variable number of molecules of glucose, and the average molecular weight is about 16,000 daltons.¹¹⁷ It is available commercially as a 7.5% solution with lactate as a buffer, calcium concentration of 3.5 mEq/L, and pH of 5.4. The solution is isoosmotic to normal plasma and icodextrin exerts oncotic pressure across the peritoneum to induce ultrafiltration through the small pores, but not through aquaporins.¹¹⁸ Given the large molecular weight, icodextrin is not absorbed across the peritoneal barrier but is removed slowly from the peritoneal cavity via the lymphatics. Only about one third of the icodextrin is absorbed, on average, over 12 hours. This, in turn, maintains the oncotic pressure for ultrafiltration for longer dwell periods and is optimally suited for use during either the overnight dwell of patients treated with continuous ambulatory PD or a day dwell for patients undergoing nighttime automated PD.¹¹⁹ The icodextrin that is absorbed is metabolized by circulating amylase into oligosaccharides of varying length, with maltose being the dominant circulating compound in patients treated with icodextrin-based dialysate.¹²⁰ Maltose readily enters the cells and is metabolized by maltase found in lysozymes of many cells into glucose.

A large number of clinical trials have tested the efficacy of icodextrin.^{121–128} In patients with high-average or high peritoneal membrane transport characteristics, icodextrin generates greater ultrafiltration volume than either 2.5% or 4.25% dextrose during the long dwells of either continuous ambulatory or automated PD, an advantage that persists

during episodes of peritonitis.^{121,122,126,129} Relative to 2.5% or 4.25% dextrose, the use of icodextrin is associated with higher ultrafiltration efficiency, i.e., a larger volume per gram of absorbed carbohydrate.^{121,126} The greater ultrafiltration volume results in a reduction in total body water and a higher likelihood of euvolemia than patients treated with dextrose-based solutions.¹²⁷ Perhaps as a result of achieving euvolemia, patients undergoing continuous ambulatory PD treated with icodextrin have been shown to have regression left ventricular hypertrophy compared to 1.5% dextrose.¹²³ Despite a reduction in total body water and claims to the contrary, there appears to be no meaningful effect of treatment with icodextrin on RKF.^{130,131}

In addition to improved control of hypervolemia, there is evidence that the use of icodextrin can exert beneficial metabolic effects, including enhanced insulin sensitivity, glycemic control and correction or attenuation of dyslipidemia.^{132,133} An observational study also suggested potentially beneficial effects on maintaining peritoneal membrane function (i.e., maintenance of the peritoneal solute transport rate) when compared with conventional PD solutions.¹³⁴ Moreover, a small clinical trial has demonstrated that patients with diabetes mellitus treated with icodextrin have a lower probability for transfer to in-center HD.¹²⁵ Finally, an observational study has shown a lower risk for death in patients treated with icodextrin.¹³⁵

Based on the accumulated body of evidence, it is recommended that icodextrin be used as a single exchange during the long dwell in patients with inadequate ultrafiltration with dextrose-based solutions; a phenomenon most likely to be present in individuals with high-average and high transport. At least two studies have examined the use of two daily exchanges with icodextrin and demonstrated a higher ultrafiltration volume and improvement in other clinically relevant parameters.^{136,137} However, such use remains investigational. Hybrid solutions containing both icodextrin and glucose have also been tested and shown to have substantially higher ultrafiltration volumes; however, none such hybrid solutions are commercially available.¹³⁸

Skin rash is the most commonly reported adverse effect with the use of icodextrin.^{122,130} The rash is typically an exfoliating rash that involves the palms and soles, is reported to occur 2 to 3 weeks from the start of exposure to the solution, and resolves after cessation of exposure. In addition, glucometers that use glucose dehydrogenase pyrroloquinolinequinone overestimate the blood glucose levels secondary to the accumulation of maltose.¹³⁹ Extreme caution should be exercised to ensure that insulin or other hypoglycemic agents are not administered because of a falsely elevated glucose test. Episodes of aseptic peritonitis have also been reported in patients treated with icodextrin. These episodes have been traced to the presence of peptidoglycan in the solution from hydrolysis of corn starch; the problem has been substantially minimized with change in manufacturing processes such that the commercially available solution has no detectable peptidoglycan.¹⁴⁰

LOW GLUCOSE-DEGRADATION PRODUCT GLUCOSE-BASED SOLUTIONS

This dual-chamber PD solution, commercially available in some parts of the world, comprises glucose as the osmotic

agent, lactate as the buffer, but has a physiologic pH and undetectable levels of glucose degradation products. Several clinical trials have examined the potential clinical benefits with such biocompatible solutions but results are mixed.^{141–152}

The primary impetus for the development of these solutions were laboratory studies that demonstrated that glucose-degradation products contributed to damage to the integrity the long-term health of the peritoneum and precluded long-term treatment with PD. Post hoc analyses of data from the BalANZ study suggests that the peritoneal solute transfer rate does not change over time in patients treated with these solutions compared to a progressive increase in peritoneal solute transfer rate with an associated decrease in ultrafiltration capacity in patients treated with conventional PD solutions.¹⁵³ However, evidence to date has not shown that these effects translate into a clinically meaningful benefit such as a lower likelihood of modality failure (i.e., transfer to in-center hemodialysis).¹⁵¹

An early crossover study demonstrated that patients had a higher urine volume when treated with a biocompatible PD solution, which decreased upon restarting treatment with conventional PD solutions.¹⁵⁴ Even though some have proposed a hemodynamic effect of such solutions with lower ultrafiltration volume and consequent volume expansion leading to natriuresis, the mechanism for higher urine volume with biocompatible PD solutions is currently not known.¹⁵⁵ Results of studies examining the effects of these solutions on RKF are heterogenous, but the totality of evidence suggests that these solutions result in a higher urine volume and a slower decline in RKF, particularly with long-term treatment.^{151,156}

At least one large clinical trial has shown a significantly lower risk for peritonitis in patients treated with a PD solution with low concentration of glucose degradation products¹⁴⁷; however, this finding has not been validated in other clinical trials.^{149,152,157–159} While observational studies have shown a lower risk for death in patients treated with biocompatible PD solutions, no such benefit was evident in a systematic review of clinical trials,^{151,160} raising the possibility of publication bias, and equally as likely, confounding by indication. Nevertheless, there is no evidence for harm with the use of these solutions and the decision to use them is driven by availability and cost.

BICARBONATE-BASED PD SOLUTION

This dual-chamber PD solution commercially available in some parts of the world contains glucose as the osmotic agent and bicarbonate as the buffer, has a physiologic pH, and has low concentrations of glucose degradation products. The development of this solution was also driven by the laboratory evidence demonstrating risk to the health and integrity of the peritoneum with the use of bio-incompatible PD solutions. There is no evidence for better preservation of structural or functional integrity of the peritoneum in humans with bicarbonate-based solutions. However, these solutions result in a more complete correction of metabolic acidosis and reduce infusion pain when compared to conventional solutions.^{161,162} A single observational study demonstrated a lower risk for death in patients treated with this solution¹³⁵; however, such a finding has not been demonstrated in a prospective clinical trial.

GLUCOSE-SPARING REGIMENS

A large number of studies have raised concern about the local and systemic effects of glucose in patients treated with PD.¹⁶³ Thus, many studies have examined the safety and efficacy of PD regimens that systematically reduce exposure to glucose with PD solutions through the course of the day. Central to such glucose-sparing regimens is substitution of one glucose-based exchange with icodextrin for the long dwell. The glucose exposure can be further reduced by substituting a second glucose-based exchange with amino acid solution. Such glucose-sparing regimens have been shown to result in improvement in glycemic control in individuals with diabetes treated with PD along with improvement in lipid parameters.^{128,132,164–166} However, there is a higher risk for adverse events in patients treated with these regimens,¹³² thought to result from a greater risk of volume overload with reduction in glucose exposure. Moreover, it remains unclear if glucose-sparing regimens reduce cardiovascular risk in patients treated with PD.¹⁶⁷ Given the higher cost of glucose-sparing regimens, the decision to use them should be made on a case-by-case basis.

PERITONEAL DIALYSIS MODALITIES

PD allows flexibility in regimen such that the prescription can be individualized to meet the lifestyle considerations of each patient while reducing the burden of treatment. Broadly speaking, PD prescriptions can be (1) continuous or intermittent, and (2) involve manual exchanges or be automated using a cycler. Continuous regimens are the ones in which there is intraperitoneal dialysate 24 hours a day, 7 days a week. Continuous regimens may involve only manual exchanges [continuous ambulatory PD (CAPD) and continuous cyclic peritoneal dialysis (CCPD)]. The most widely used intermittent regimens comprise the use of a cycler to deliver PD for a part of the day – either at night (nocturnal intermittent PD; NIPD), or day (diurnal intermittent PD). On occasion, patients doing CAPD may be prescribed a “dry night”. Such intermittent regimens are appropriate only for individuals with significant RKF.¹⁶⁸ Rarely, intermittent PD is delivered by performing frequent exchanges over 10 to 36 hours every few days. This is used either as a bridge for individuals in the immediate postoperative period (such as “urgent start” PD) or as palliative therapy for patients as part of end-of-life care.^{169,170}

Arguably, patients with slower peritoneal solute transfer rates (slow or low transporters) would require longer dwell periods to achieve adequate solute removal. In contrast, patients with faster peritoneal solute transfer rates (fast or high transporters) could benefit with shorter nighttime dwells to optimize daily ultrafiltration. However, most patients can be successfully treated with either continuous ambulatory or automated PD, particularly in the presence of RKF.

A large number of observational studies have examined the potential effects of treatment with continuous ambulatory or automated PD on clinically meaningful outcomes; the evidence has recently been summarized in a narrative review.¹⁷¹ In the early days of PD, use of the cycler reduced the number of connections and disconnections that needed to be made during the course of the day when compared to patients

treated with continuous ambulatory PD. Studies from this period showed a lower risk for peritonitis in patients treated with automated PD. However, there has been a widespread adoption of flush-before-fill, twin-bag systems and disconnect systems for automated PD such that in contemporary practice, there is no difference in risk for peritonitis in patients treated with continuous ambulatory or automated PD.¹⁷¹

Concern has been raised that the total daily sodium and water removal may not be adequate in patients treated with automated PD.^{172–175} This phenomenon has been attributed to sodium sieving during short nighttime dwells and fluid reabsorption during the long day dwell that are typical for automated PD prescriptions. Sodium sieving refers to the exclusion of sodium when water is transported across ultra-small aquaporin pores, leading to disproportionately less sodium removal. This occurs early in the dwell, but is compensated by diffusive equilibration of plasma sodium with the dialysate, if the dwell continues for long enough. However, comparative studies of CAPD with automated PD have typically not taken into account overfill for flush-before-fill that is common in bags used for continuous ambulatory PD.¹⁷⁶ As a result, these studies may have overestimated the sodium and water removal with continuous ambulatory PD. This explanation is consistent with the observed results in comparative studies, showing no difference in the prevalence of volume overload or hypertension control in patients treated with continuous ambulatory or automated PD.¹⁷⁷

Despite a reduction in burden of treatment, there is no evidence that patients treated with automated PD have lower mortality, enhanced health-related quality of life, or lower risk for transfer to in-center hemodialysis.^{178–182} The differential use of the two PD modalities is likely to continue to be driven by lifestyle and cost considerations with a likely greater use of automated PD in many countries such as the United States or Western Europe.

DIALYSIS ADEQUACY

The concept of an “adequate” PD dialysis dose has been strongly influenced by randomized controlled trials and better understanding of the multiplicity of factors affecting PD patients with little or no RKF.^{183–186} Before the Adequacy of Peritoneal Dialysis in Mexico (ADEMEX) study and the HD Study (HEMO), the notion that prevailed was that a continuous increase in small solute clearances would lead to better clinical results and patient survival.^{183,184,187} The present concept of an adequate dialysis now encompasses attention to a multiplicity of clinical and nonclinical variables (multi-dimensional concept), including RKF, cardiovascular effects of blood pressure control, ultrafiltration and volume overload, mineral metabolism, nutrition, and individual psychologic and nonphysiologic (i.e., those related to other domains of health status) indicators (Table 64.3).^{183,187–189}

Adequate dialysis is defined as the administration of an effective dosage of dialysis solution, capable of keeping a patient clinically asymptomatic and active and maintaining sufficient correction of the altered metabolic and homeostatic components secondary to the loss of kidney function.¹⁸⁸

Optimal dialysis is defined as either (1) the dose capable of reducing morbidity and mortality associated with ESKD and with the dialytic procedure itself or (2) the dose above

Table 64.3 Considerations for Adequate Dialysis

Clinical manifestations: fluid balance, systemic blood pressure control, and cardiovascular risk
 Residual kidney function
 Acid-base homeostasis
 Nutritional status
 Calcium-phosphorous metabolism homeostasis
 Inflammation
 Small solute clearance
 Middle molecule clearance
 Psychologic and quality-of-life indicators

Table 64.4 Calculations for Dosage of Dialysis Solution

Fractional Urea Clearance (Kt/V_{urea})
 Daily total Kt/V_{urea} = peritoneal Kt/V_{urea} + renal Kt/V_{urea}
 Weekly Kt/V_{urea} = $7 \times (((24\text{-hr urea D/P} \times 24\text{-hr EV})/V) + [(U_{\text{urea}}/P_{\text{urea}}) \times 24\text{-hr UV}])$
 Total Body Water (Watson's Formula)^a
 Male: $V = 2.447 - (0.3362 \times \text{weight}) - (0.1074 \times \text{height}) - (0.09516 \times \text{age})$
 Female: $V = -2.097 + (0.2466 \times \text{weight}) + (0.1069 \times \text{height})$

^aIn Watson's formula, age is in years, height is in centimeters, and weight is in kilograms.

which an increase does not justify the increased burden of treatment.^{190–192}

INDICATORS TO EVALUATE DIALYSIS ADEQUACY

The dialysis dose prescribed is often based on clinical evaluations as the presence or absence of symptoms related to uremia (e.g., nausea, vomiting, dysgeusia, sleeping disorders). Nevertheless, this may be highly subjective and prone to dosage underestimation and while clinical indicators should not be ignored, therapeutic decisions should not be based solely on them.¹⁸⁶ Clearance of urea is at present still employed as a marker of dialysis dose, in conjunction with other indicators described above. Residual kidney function tends to be maintained in PD for a longer period than in HD and may account for a significant proportion of the total clearance.^{191,193,194}

Urea is the traditional solute to quantitate dialysis as its concentration is increased in CKD and has a low molecular weight (60 kDa), which allows rapid diffusion between body compartments and therefore application of a single-pool model for approximation. In addition, its volume of distribution is the total body water, it easily diffuses across the dialysis membrane, and is easy to measure.^{188–194} After the National Cooperative Dialysis Study (NCDS) results were published in the early 1980s, the urea kinetic model took on a prominent role in dosage prescription in PD as well, notwithstanding the fact that it was created for HD and that its application to PD was accomplished only with inferences and analogies.¹⁹⁵

The fractional clearance of urea is expressed as Kt/V_{urea} , which is the clearance of urea (K) per time unit (t) in relation to its volume of distribution or total body water (V). Peritoneal Kt is calculated by collecting a 24-hour amount of effluent dialysate and determining its urea concentration; this in turn is divided by the plasma urea concentration (D/P_{urea}). To compare clearance values among patients, these values are normalized to a function of patient size: For urea, the metric is typically the volume of urea distribution (V). In PD practice, Kt/V_{urea} is expressed as a total (sum of peritoneal pKt/V_{urea} and renal urea clearances). Renal Kt is calculated similarly, with a 24-hour urine collection. Kt/V_{urea} may be expressed as a daily value but is usually multiplied by 7 and expressed as a weekly value. This was done to compare PD and HD delivered dosages, yet these comparisons are not logical, given the intermittent nature of HD and the continuous

Table 64.5 Determinants of Solute Clearance and Fluid Removal in Peritoneal Dialysis

Patient-Related Factors	Prescription-Related Factors
Residual kidney function	Dialytic modality
Body mass index	Frequency of exchanges per day
Peritoneal transport type	Volume of each exchange
	Dwell time
	Dialysis fluid tonicity

nature of PD (Table 64.4).¹⁹⁶ The urea kinetic model, repeatedly validated for HD prescription, has been applied to PD, but its validity is questionable. If the same dosing principles employed for HD were used for CAPD, then patients on PD would be considered grossly underdialyzed because their Kt/V_{urea} values would be much lower in absolute numbers; nevertheless, most patients on PD do not have more uremic manifestations, higher rates of morbidity, or mortality than do patients on HD.

DETERMINANTS OF SOLUTE CLEARANCE AND FLUID REMOVAL IN PERITONEAL DIALYSIS

The initial PD prescription is defined empirically; a series of factors that influence clearance of solutes and fluid removal are considered. Some of these factors are modifiable and others are nonmodifiable (Table 64.5).^{192,197,198}

PATIENT-RELATED FACTORS

Residual kidney function contributes significantly to fluid removal and variably to solute clearance in PD patients. In these patients, RKF is best estimated by measuring the weekly renal component of the Kt/V_{urea} . Each milliliter per minute of urea clearance accounts for an additional 0.25 L to the total weekly Kt/V_{urea} .¹⁹⁹ Estimation of RKF enables physicians to plan the PD prescription and during follow-up, RKF should be evaluated periodically as it tends to decrease over time; thus, the dialysis prescription should be modified to compensate. A faster decline in RKF is associated with lower survival in PD patients.²⁰⁰

VOLUME OF DISTRIBUTION AND BODY SURFACE AREA

The volume of distribution of urea is equivalent to the total body water, which can be estimated by the Watson or Hume formula, but these formulas may underestimate total body water in patients on dialysis and potentially overestimate dialysate dosage.²⁰¹ An equation has specifically been developed for patients undergoing maintenance dialysis and has been validated to provide superior prediction of total body water in HD and, potentially, also in patients on PD.²⁰²

PERITONEAL TRANSPORT TYPE

Fig. 64.2 displays the D/P_{Cr} versus dwell time of a dialytic exchange in patients with high and low (or fast and slow) peritoneal transport. The D/P_{Cr} equilibrium (ratio of 1.0) is achieved earlier in high peritoneal transport; therefore, a shorter dwell time is preferred for patients with higher peritoneal transport rates, and a longer dwell time is required to augment solute clearance in patients with lower transport rates (Fig. 64.2). In comparison, patients with lower peritoneal transport rates achieve higher ultrafiltration volumes, whereas patients with higher peritoneal transport rates reabsorb larger quantities of glucose, an occurrence that reduces the gradient for ultrafiltration and generates net fluid reabsorption with longer dwell times (Fig. 64.3).⁴⁴ Patients with higher peritoneal transport rates obtain better ultrafiltration and adequate clearance of small molecules with techniques that promote shorter dwell times, such as APD, prescribed with multiple short exchanges. In contrast, patients with lower peritoneal transport rates benefit from longer exchange dwell times to augment removal of small molecules while preserving the ultrafiltration capability. It is noteworthy that most patients have intermediate transport rates; therefore, individualized evaluation is required to prescribe the best regimen.⁴⁴

In addition to the discussed relation between transport type with ultrafiltration and solute clearance, is important to consider that the smaller the solute, the faster is the diffusive equilibrium reached. Urea, with a molecular weight of 60 Da, reaches equilibrium much faster than does creatinine

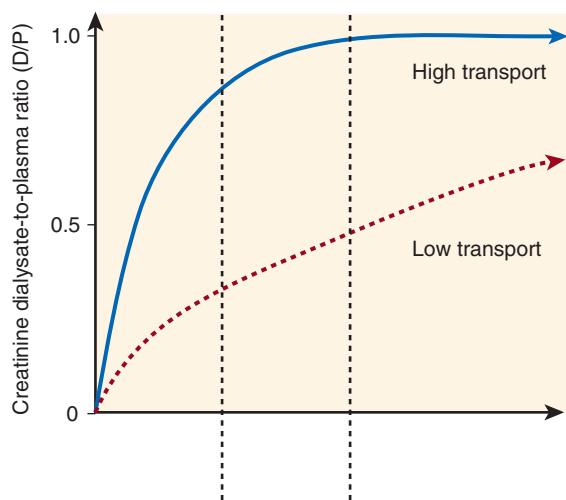


Fig. 64.2 The ratio of dialysate-to-plasma ratio for creatinine (D/P_{Cr}) plotted against dwell time of a dialytic exchange in conditions of high and low peritoneal transport.

(molecular weight of 112 Da). Conditions influencing small-molecule solute clearance may not be relevant to clearance of large molecules, charged, or protein-bound solutes. A more detailed discussion of uremic solutes and the limitations of using urea as a solute marker are provided in Chapter 52.

PRESCRIPTION-RELATED FACTORS

The main modifiable factors in the PD prescription are: frequency of exchanges, volume of PD fluid exchanges, and tonicity of solutions.

Frequency of Exchanges

Traditional CAPD prescription is based on four exchanges of 1.5–3.0 L every day. Yet some patients with small body surface area or significant RKF may start their treatment with three or fewer exchanges per day; this approach has been defined as incremental, as with time patients are expected to require an increase in dosage.^{203,204} As creatinine is a larger molecule than urea, an increase in exchange frequency is less effective than an increase in fluid volume per exchange for removal of creatinine, particularly in patients with low peritoneal transport rates.¹⁹⁹ In APD, an increase in frequency of exchanges also induces an increase in solute clearance as the concentration gradient between dialysate and blood is maximized, but an increase in exchange frequency obligates a larger proportion of the total treatment time be dedicated to infusion and drainage. Therefore, there is a point at which the number of exchanges may be counterproductive in relation to clearance attained (and in relation to costs). The “break even” point is variable from patient to patient and is related in part to the peritoneal transport rate.¹⁹⁷ A recent addition to dialysis prescription that may add important benefits is the possibility of remote monitoring of dialysis performance in patients with APD, which may help to personalize, and thus improve individual dialysis prescription.²⁰⁵

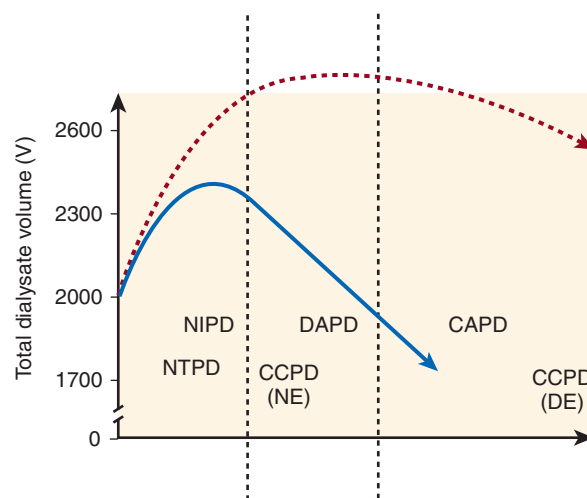


Fig. 64.3 Differences in ultrafiltration and net fluid reabsorption between patients with lower peritoneal transport (red line) and patients with higher peritoneal transport (blue line). CAPD, Continuous ambulatory peritoneal dialysis; CCPD, continuous cycling peritoneal dialysis; DAPD, daytime ambulatory peritoneal dialysis; DE, daytime exchanges; NE, nocturnal exchanges; NIPD, nocturnal intermittent peritoneal dialysis; NTPD, nightly tidal peritoneal dialysis.

The best way to increase clearance in NIPD is to add a daytime exchange. The daytime exchange increases Kt/V_{urea} by approximately 25%. The main drawback is that a long daytime exchange often leads to net fluid reabsorption, particularly in patients with high and high-average peritoneal transport rates. A way to avoid this problem is to individually tailor the daily exchange time for each patient according to the clinical response. In some patients, additional clearance may be obtained by adding a second or third daytime exchange or by switching to CAPD.

Increase in Volume Exchange

An increase in fill volume is generally a more effective means of increasing small-molecule solute clearance than is an increase in exchange frequency. Therefore, it is one of the main prescription actions to achieve goals for CAPD, as well as for APD. Volume exchange is limited by patient body surface area and tolerance of higher fill volumes, and complications such as PD fluid leaks, and hernias of the abdominal wall.^{192,197,198}

Increase in Solution Osmolality

A dialysate with a higher tonicity induces an increase in ultrafiltration volume, and is employed mainly to avoid or control hypervolemia, often present in patients without RKF independent of their transport type, and in patients with high and high-average transport rates and in nonadherent patients that ingest large amounts of salt or water. These dialysates may also induce a convection-driven, quantifiable increase in solute clearance. It is important to point out that use of hypertonic glucose solutions has potential adverse metabolic consequences, as addressed below. The use of polyglucose solutions (icodextrin) allows for increased ultrafiltration with a lower risk of inducing metabolic complications.¹⁹⁷

CLEARANCE TARGETS AND CLINICAL OUTCOMES

When outcomes of patients on PD were assessed in the prospective Canada-USA cohort study of adequacy in PD (CANUSA), there was an inverse correlation between Kt/V_{urea} and mortality.¹⁹³ This finding led to the implementation of “evidence-based” recommendations by the Kidney Disease Outcomes Quality Initiative (KDOQI) that the weekly dose of CAPD be at least Kt/V_{urea} of 2.0 for all patients and that the total creatinine clearance rate be at least 60 L/week/1.73 m² for high and high-average transporters and 50 L/week/1.73 m² for low and low-average transporters. In a reanalysis of CANUSA, the same authors noted that one conclusion was erroneous.²⁰⁶ In CANUSA’s first analysis, the equivalence of renal and peritoneal clearances was assumed, and therefore they were merely added. For every 5-L/week/1.73 m² higher residual creatinine clearance, there was a 12% lower relative risk of death; no such association with peritoneal creatinine clearance was evident. The observed survival advantage associated with higher small solute clearances could be accounted for by RKF.²⁰⁶ This fact was further supported by the findings of other investigators who have stressed the greater importance of RKF. Other studies were unsuccessful in evaluating the effect of peritoneal clearance on patients.^{207–209}

The ADEMEX study was a prospective, randomized clinical trial to evaluate the effect of higher peritoneal clearance on patient survival.¹⁸³ The trial included 960 patients with incident and prevalent kidney disease (>50% anuric) who were receiving CAPD with four 2-L exchanges, and had peritoneal creatinine clearance rates of less than 60 L/week/1.73 m² at 24 centers in Mexico. Trial participants were monitored for more than 2 years. A control group received a standard prescription (four 2-L exchanges), and for the experimental group the prescription was changed to achieve a peritoneal creatinine clearance of 60 L/week, by increasing dwell volumes and, when necessary, adding a fifth automated night exchange. The control group achieved a weekly peritoneal creatinine clearance rate of 46 L and a peritoneal Kt/V_{urea} of 1.62, whereas the experimental group achieved a peritoneal creatinine clearance rate of 56 L and a Kt/V_{urea} of 2.13 per week; thus, substantial group separation was achieved. No differences were observed between the two groups in primary (risk of death) and secondary (technique failure, hospitalization, nutritional status) outcomes.¹⁸³ These results support the hypothesis that increases in creatinine and urea clearance with higher PD dosage, within the studied ranges, do not enhance patient survival.

Data from several subgroups—including younger and older patients, those with or without RKF, those with or without diabetes, and those with larger and smaller body surface areas—were analyzed and, again, showed no differences in mortality or other outcomes by randomized group among these subgroups. Control participants were more likely to drop out of the study due to uremia. The results of the study may not be fully generalizable as patients in the ADEMEX trial were smaller in size, younger, and had more pronounced protein-energy wasting than other PD populations.²¹⁰ However, these arguments do not preclude the external validity of the trial, inasmuch as comorbid conditions, general survival rates, and causes of mortality were similar to those observed in all other studies, including CANUSA.¹⁹³

Results of ADEMEX were supported by those of Li and colleagues in a randomized clinical trial of 320 patients with incident kidney disease who were receiving CAPD and whose baseline renal Kt/V_{urea} values were less than 1.0.¹⁸⁵ Patients were randomly assigned to three target groups: In group A, Kt/V_{urea} ranged from 1.5 to 1.7; in group B, Kt/V_{urea} ranged from 1.7 to 2.0; and in group C, Kt/V_{urea} exceeded 2.0. Total Kt/V_{urea} values of the three groups were significantly different, and this difference was mostly attributable to peritoneal Kt/V_{urea} . There was no difference in patient survival, serum albumin concentration, and hospitalization rates among the three groups; however, more patients from group A required erythropoietin and were withdrawn from the study by their physicians. Thus, patients with total Kt/V_{urea} values below 1.7 had more anemia and clinical complications, and yet there was no difference in survival or other outcomes for groups B and C; i.e., patients targeted to Kt/V_{urea} between 1.7 and 2.0 or above 2.0. Results from these two clinical trials support the notion that a total Kt/V_{urea} of 1.7 or higher is an appropriate target for small solute clearance in PD.¹⁸⁵ In addition, Lo and colleagues performed a 10-year retrospective survival analysis of 150 anuric patients on PD, according to the baseline peritoneal Kt/V_{urea} at time of documentation of anuria and at the latest PD prescription (based on Kt/V_{urea}).²¹¹ Baseline Kt/V_{urea} was not independently associated with patient survival;

Table 64.6 Adequacy Targets in Peritoneal Dialysis

Index Measure	Adequacy Index		
	European Renal Association (2005)	Kidney Disease Dialysis Outcomes Quality Initiative (2006)	International Society of Peritoneal Dialysis (2006)
Total Kt/V_{urea}			
CAPD	1.7	1.7	1.7
APD	1.7	1.7	1.7

nevertheless, patients with peritoneal Kt/V_{urea} values lower than 1.67 had poorer survival rates. Survival rates did not differ among patients with Kt/V_{urea} either higher or lower than 1.80. The survival rate was best among female patients with Kt/V_{urea} of 1.67 to 1.86, followed by those with Kt/V_{urea} higher than 1.86, and was lowest in those with Kt/V_{urea} lower than 1.67. Peritoneal Kt/V_{urea} below 1.67 was consistently associated with lower survival rates. These observational results could be confounded by protein energy wasting, with “shrinkage” of the total body water (largely housed within skeletal muscle) space. Parallel controlled trials of dialysis dose in APD are lacking.

Given the current body of evidence, clinical practice guidelines recommend that Kt/V_{urea} is still the best available index of “adequacy” and that a value of total Kt/V_{urea} of 1.7 or more can be considered adequate (Table 64.6).^{182,186,195,198,212} Overemphasis on urea kinetic modeling has resulted in the neglect of several relevant issues. Chronic overhydration has major negative consequences, and cardiovascular disease is the most frequent cause of morbidity and mortality in patients with ESKD in general and in those on PD specifically.²¹² The adverse effects of overhydration may counterbalance the marginal benefits of higher solute clearance if the latter is solely attended to in lieu of the former. Salt and water overload have to be closely monitored and corrected through dietary restrictions, ultrafiltration, pharmacologic interventions (i.e., use of oral diuretics in patients with RKF), or a combination thereof.

Renal clearance is not equivalent to peritoneal clearance, inasmuch as the kidney provides benefits that are not achieved with PD, including better middle molecule clearance, volume control, and metabolic and endocrine benefits. Hyperphosphatemia and other abnormalities of mineral metabolism are strongly associated with cardiovascular morbidity and mortality.²¹³ Protein-energy wasting should also receive close attention because it is present in many patients with CKD (see Chapter 60). Protein-energy wasting is a negative prognostic indicator to be considered even before the need of dialysis and is discussed in the “Nutritional Counseling and Nutrient Supplements” section. Attention to nontraditional risk factors, particularly inflammation, almost always present in patients with kidney disease, is important; inflammation is also discussed in more detail in the “Inflammation and Peritoneal Dialysis” section. Social and psychologic factors are also important for the care of the patient on PD.

Volume status constitutes an important determinant for PD patient survival and for an adequate prescription. Volume overload has an impact on cardiovascular morbidity as well

as mortality in PD patients. Therefore, it is of upmost importance to consider normovolemia (achieved via “adequate” ultrafiltration) as a major target of dialysis adequacy.^{192,198,214} Determination of optimal volume status is a complex issue and recent studies with bioimpedance spectroscopy have proven useful for this purpose.^{215,216}

NUTRITION AND PERITONEAL DIALYSIS

Protein-energy wasting is often present in patients on PD, and correlates directly with time on PD.^{217–219} A deficient nutritional status in the PD patient is strongly associated with morbidity and mortality.^{220–222} Multiple variables are employed to assess nutritional status, including serum albumin and prealbumin concentrations, body cell mass, total body nitrogen, creatinine excretion, anthropometric indices, subjective global assessment (which ironically, is neither fully subjective nor global), bioimpedance vector analysis, and composite nutritional scores. In order to better evaluate and monitor the nutritional status of a patients receiving PD, the use of multiple tools is required, as no single measure provides a complete evaluation.^{223,224}

The cause of protein-energy wasting in PD patients is multifactorial and complex (Table 64.7). Nutrient losses during dialysis, low nutrient intake, comorbid conditions, chronic inflammation, metabolic acidosis, loss of RKF, uncorrected uremia, peritoneal transport type, aging, and a variety of endocrine disorders contribute to the deficient nutritional status.^{225–229} Some studies have shown that patients with high peritoneal membrane permeability could be at higher risk for protein-energy wasting; however, studies examining the links between peritoneal transport and nutritional status have yielded contradictory results.^{229–231} It has been long described that patients on dialysis may have at least two different types of protein-energy wasting.²³² The first one related to low nutrient intake and a second form associated with inflammation and cardiovascular disease, not unlike the classic definitions of marasmus and kwashiorkor; the latter form is more common in PD, as the modality often contributes to the delivery of adequate (or sometimes in conjunction with intake, excessive) calories. These types of protein-energy wasting often coexist in the clinical setting.

A diversity of therapeutic strategies to treat or prevent declining nutritional status in the PD patient has been suggested. These include: nutritional counseling for adequate nutrient intake, treatment of reversible causes of anorexia, and correction of catabolic factors (inflammation, correctable comorbidities, uremia, and acidosis; see Table 64.7).

Table 64.7 Etiology of Protein-Energy Wasting in Patients on Peritoneal Dialysis

Causes of Protein-Energy Wasting	Possible Management Strategies
Nutrient Losses	
Amino acids	Nutritional counseling
Peptides	30-35 kcal/kg/day
Proteins	1.2 g of protein/kg/day
Water-soluble vitamins	Nutrient and vitamin supplements
Other bioactive compounds	Dialysis solutions with amino acids
Low Nutrient Intake	
Anorexia	Treatment of reversible causes of anorexia
Impaired gastric emptying	Increasing the dosage of dialysis solution
Altered taste sensation	Prokinetic agents
Unpalatable diet	Nutritional counseling
Inadequate clearance of anorexigens	30-35 kcal/kg/day
Intercurrent illness	1.2 g of protein/kg/day
Emotional distress	Nutrient and vitamin supplements
Impaired ability to procure, prepare, or ingest food	Dialysis solutions with amino acids
Comorbidity	Treatment of comorbid illnesses
Chronic Inflammation	Extracellular fluid volume control Antiinflammatory drugs (in the future)
Metabolic Acidosis	Correction of acidosis
Loss of Residual Kidney Function	Renin angiotensin system inhibitors High-dose furosemide (urine volume) Avoidance of nephrotoxins
Possible Role	
Endocrine disorders of uremia	Recombinant growth hormone
Fast peritoneal membrane transport	Correction of anemia and comorbid conditions
Occult gastrointestinal bleeding	

NUTRITIONAL COUNSELING AND NUTRIENT SUPPLEMENTS

Nutritional counseling is the first-line intervention to achieve adequate nutrient intake.²³³ The usual recommended daily energy intake for patients on maintenance PD is 35 kcal per kilogram of body weight per day for those younger than 60 years of age and 30 to 35 kcal/kg/day for those aged 60 years or older.²²³ Nevertheless, the presence of obesity may modify our prescription accordingly, as in some patients a lower intake is preferred in order to limit weight gain or even to induce weight loss. Energy intake in patients on PD, in addition to dietary intake, should take into account glucose absorbed from the dialysate, which is dependent on several factors such as peritoneal transport, volume, dwell time, and

glucose concentration. Energy from dialysate on average is around 20% of the total energy intake, corresponding to 3 to 13 kcal/kg/day.²³⁴ In relation to protein intake, no less than 1.2 g/kg ($\geq 50\%$ of proteins with high biologic value) is the daily recommendation for adults on PD, and in some specific instances of severe protein-energy wasting it should be higher²³³; nevertheless, it is important to point out there is a significant lack of evidence from randomized trials to confirm optimal protein intake in this population.²³⁵ It is important to point out that actual protein and energy intake of PD patients is often considerably lower than recommended and anorexia, dysgeusia, and gastrointestinal symptoms could be contributing to this situation.^{233,236}

Intensive nutritional support may be of value during hypercatabolic conditions, including episodes of peritonitis. However, the effect of counseling on the nutritional status of affected patients remains largely untested. One report suggested that nutritional counseling, as an isolated measure, may maintain nutritional status despite a decrease in RKF and higher rates of systemic inflammation.²³⁷ Administration of oral nutritional supplements may contribute to improvement of nutritional status. Few controlled clinical trials have involved commercially available supplements, dry egg albumin-based supplements, and whey-protein supplements as a nutritional intervention, with results that are positive, suggesting the benefit of this approach.²³⁸⁻²⁴³

DIALYSIS SOLUTIONS WITH AMINO ACIDS

Management of protein-energy wasting with the available PD solutions has been poorly explored and seems to require a personalized prescription.²⁴⁴ The use of amino acids has been explored and in few randomized trials conducted to date, anthropometric and biochemical measures have been better preserved in some but not all treated patients.^{245,246}

REVERSIBLE CAUSES OF ANOREXIA

Increasing dialysis dose to correct anorexia is an approach often attempted but found to be ineffective in large randomized trials.^{183,209,217} Gastroparesis is a frequent complication in patients on PD, particularly, but not exclusively, among those with diabetes²⁴⁷; correction of gastroparesis may increase dietary intake and reduce nausea and vomiting. The use of prokinetic agents has been reported to increase serum albumin concentration in hypoalbuminemic patients on dialysis with delayed gastric emptying.²⁴⁸

METABOLIC ACIDOSIS

Given that acidosis induces protein catabolism, correction of acidosis with oral alkali has been a measure to improve nutritional status (see Chapter 60). In a randomized trial conducted among patients on PD, oral sodium bicarbonate supplementation resulted in a significant increase in the plasma bicarbonate concentration, along with improvements in some anthropometric measurements and nutritional status (evaluated by subjective global assessment); this in turn was associated with shorter hospitalization period and reduced morbidity.²⁴⁹ A recent study demonstrated that oral bicarbonate administration may decelerate the loss of RKF in PD patients.²⁵⁰

ANABOLIC HORMONES

An anabolic strategy is to administer recombinant growth hormone. Its administration has been reported to be effective in improving nutritional parameters in short-term studies in children; however, its use is limited by cost and the development of hyperglycemia and other side effects.²⁵¹ Similar results have been observed with the use of insulin-like growth factor-I.²⁵² In addition, in small randomized trials, androgenic (anabolic) steroids have been shown to improve some parameters of nutritional status, yet significant risks and side effects preclude their use in clinical practice.²⁵³ Ghrelin is a hormone synthesized by gastric endocrine cells and is involved in regulation of food ingestion and energy metabolism. Patients on PD have lower ghrelin concentrations than patients on HD or controls.²⁵⁴ Although ghrelin, administered subcutaneously, enhances short-term food intake in patients with mild to moderate protein-energy wasting, further studies are necessary to determine the therapeutic role of this hormone.²⁵⁵ Other hormones and cytokines such as leptin, TNF α , and IL-6, all of which have been found to be elevated in patients undergoing maintenance dialysis, could participate in the loss of appetite present in patients on PD.²⁵⁶

L-CARNITINE

Advanced chronic kidney disease is associated with abnormal L-carnitine metabolism. At present, data demonstrating a beneficial effect of L-carnitine on nutritional status in PD patients are insufficient to recommend routine use.^{257–259}

INFLAMMATION AND PERITONEAL DIALYSIS

An inflammatory phenomenon evidenced by increased proinflammatory cytokines and acute phase reactants is observed in 12% to 65% of patients with advanced CKD before they start dialysis, and it is aggravated with both PD and HD.^{260,261} Systemic inflammation is strongly related to atherosclerosis and protein-energy wasting.^{260,261} Local intraperitoneal inflammation induces important structural alterations, including thickening and cubic transformation of mesothelial cells, fibrin deposition, fibrous capsule formation, perivascular bleeding, and interstitial fibrosis, and is associated with peritoneal solute transfer rate.⁷⁰ These structural alterations induce clinical and functional changes, including ultrafiltration failure, which can occur in a large proportion of longstanding PD patients.²⁶²

In the stable patient on PD without active peritonitis or other infection, several conditions may be implicated in the origin of systemic inflammation, including the use of bioincompatible dialysis solutions. Additional factors, such as rapid peritoneal membrane transport, low RKF, and overhydration, have also been associated with inflammation.²⁶³ It is clear and a universal consensus that systemic inflammation, independently or in combination with all the variables mentioned previously, is an important factor associated with morbidity and mortality.^{264,265–266}

There is a strong association among fluid overload, cardiac natriuretic hormones, and proinflammatory cytokines.²⁶⁷ Adequate control of the hypervolemia may reduce

a heightened inflammatory state, particularly in patients with high transport rates. When hypervolemia was ameliorated in high and high-average transport PD patients with NIPD, a reduction of serum C-reactive protein and IL-6 was observed.²⁶⁸

Results of small studies indicate that agents such as thiazolidinediones and statins may have relevant antiinflammatory effects, but these effects are modest.^{269,270} Angiotensin-converting enzyme inhibitors may suppress C-reactive protein and oxidized low-density lipoprotein (LDL) cholesterol in patients with diabetes, but they do not appear to reduce inflammation in patients on HD.^{271,272} In contrast, some evidence has been generated to support the idea that inhibition of the renin-angiotensin system may have an anti-inflammatory effect on the peritoneal membrane.²⁷³

At present, there are no studies testing the impact of treatment for chronic nondialysis-related infections (*Chlamydia pneumoniae*, *Helicobacter pylori*, dental-gingival infections, or viral hepatitis) on the inflammatory status of PD patients. The development of drugs targeting specific mediators of the inflammatory response, such as IL-6 or tumor necrosis factor may hold promise for the future.

RESIDUAL KIDNEY FUNCTION AND PERITONEAL DIALYSIS

Peritoneal dialysis is associated with preservation of RKF, and PD has been shown to preserve RKF longer than hemodialysis.²⁷⁴ PD may preserve RKF better than HD in part because, in comparison to HD, PD is associated with less hemodynamic instability, less frequent volume depletion, and avoidance of the extracorporeal system of hemodialysis that serves as sources of systemic inflammation, oxidative stress, and subsequent kidney injury. Several factors have been associated with loss of RKF in patients receiving PD, including female sex, diabetes mellitus, and nonwhite race, as well as comorbid conditions including congestive heart failure, poorly controlled hypertension, coronary artery disease, and aminoglycoside use.^{275–278}

Multiple cohort studies have confirmed a positive association between the presence and degree of RKF and survival in PD patients.^{206,279,280} Subsequent studies have consistently shown an association between loss of RKF and higher risk for volume overload, left ventricular hypertrophy, and congestive heart failure, all of which are meaningful to patients, and associate with survival.^{281,282} These observations of greater survival in PD patients with better preserved RKF has been postulated to result from several factors including: renal clearance of circulating proinflammatory cytokines, middle molecular weight uremic toxins and protein-bound toxins (e.g., p-cresol), which are otherwise poorly removed with dialysis alone; improved anemia management due to persistent endogenous erythropoietin production; and preserved renal salt and water excretion allowing better maintenance of euvolemia and normotension.^{281,282} Additionally, preserved RKF allows for better maintenance of normal bone mineral metabolism; RKF loss in PD patients is associated with higher serum phosphate concentrations and higher serum FGF-23 levels (independent of serum phosphorus), both of which have been implicated in causing increased arterial stiffening and valvular calcification, and hence increased cardiovascular

mortality.²⁸³ Loss of RKF per se is associated with increased valvular calcification and cardiac hypertrophy.²⁸⁴

Nutritional status, which is strongly related to the presence of inflammation, is better maintained in the presence of RKF.^{226,285} Appetite, macro- and micronutrient intake are linked to better preserved RKF.^{226,266} Moreover, anuric patients seem to have higher resting energy expenditure compared to patients with RKF, which may partially explain the superior nutritional status seen in the latter group.²⁸⁶ Better nutrition has been linked to a stronger immune system, and indeed incident PD patients with RKF have been shown to have lower rates of peritonitis and peritonitis-related mortality, compared with anuric patients.^{285,287} The strategy of preferentially using PD as the initial dialysis modality in patients with RKF is intriguing and deserves additional investigation; small studies seem to show a better survival in patients who initiate PD therapy and subsequently transfer to hemodialysis therapy compared with matched hemodialysis-only patients, particularly those who begin hemodialysis with a central venous catheter.^{288–290}

MEASURES TO PRESERVE RESIDUAL KIDNEY FUNCTION

The effect of PD modality (i.e., CAPD versus APD) on RKF preservation has been investigated in numerous studies.¹⁷¹ While some reports suggest that APD is associated with a more rapid decline of RKF than CAPD in spite of similar peritoneal UF volumes, the preponderance of evidence suggests that PD modality has no effect on change in RKF.¹⁷¹

High-dose furosemide has been shown to preserve urinary volume, increase sodium removal and decrease weight gain better than placebo, without an independent effect on solute clearances per se.²⁹¹ On the other hand, hypovolemia and hypotension have been shown to be independently associated with a more rapid loss of RKF, thus special care is needed to achieve a safe balance between extracellular fluid volume overload and hypovolemia caused by overzealous diuresis and ultrafiltration.

Biocompatible PD fluids have been hypothesized to be less glomerulotoxic than conventional glucose-based solutions; the effect of low glucose-degradation product glucose-based PD solutions and icodextrin is discussed in more detail above. In summary, evidence suggests that low-glucose degradation product glucose-based PD solutions are associated with slower decrease in urine volume and solute clearance in the long-term. In contrast, there is no meaningful effect of icodextrin on the rate of loss of residual kidney function.

Inhibitors of the renin-angiotensin-aldosterone system blockade slow the progression of chronic kidney disease of multiple etiologies, with the preponderance of evidence accumulated in type 2 diabetes. In two small trials of prevalent patients receiving PD, angiotensin converting enzyme inhibitors and angiotensin receptor blockers promote preserved RKF relative to control antihypertensive agents.^{292,293}

sudden cardiac death, and stroke—is common among patients with ESKD.²⁹⁴ Moreover, patients on PD have a remarkably high risk of dying from myocardial infarction, arrhythmias, valvular disease, and sudden cardiac death.²⁹⁵ Vascular and valvular calcification, volume overload with resultant left ventricular hypertrophy, inflammation and accelerated atherosclerosis all contribute to cardiovascular mortality in patients with ESKD.²⁹⁶

The interplay of many pathways underlie the development of cardiovascular disease in patients on PD. These pathways include traditional risk factors (diabetes mellitus, hypertension, dyslipidemia, sedentary lifestyle, left ventricular hypertrophy, smoking, male sex, insulin resistance), uremia-specific factors (anemia, phosphate retention, vascular calcification, uremic toxins, volume overload, hyperparathyroidism), novel risk factors (inflammation, oxidative stress, endothelial dysfunction, activation of the sympathetic nervous system, protein-energy wasting, carbamylation of proteins, epigenetic changes), and genetic factors.²⁹⁷ Additionally, there are factors specific to PD that increase cardiovascular risk, e.g., hypokalemia (more common with PD than HD) has been demonstrated to confer a higher risk of death in patients receiving PD²⁹⁸; similarly, overhydration and loss of RKF have each been associated with higher risk for cardiovascular disease.^{183,206,279,299–301}

MANAGEMENT OF CARDIOVASCULAR DISEASE IN PERITONEAL DIALYSIS

Hypertension is a well-recognized risk factor for cardiovascular disease in the general population. The relationship between hypertension per se and cardiovascular disease in dialysis patients is complex due to the myriad comorbidities seen in the ESKD population. While short-term studies have found a weak or even inverse relation between hypertension and cardiovascular disease in ESKD, longer-term studies have found benefit to control of hypertension.^{302–304} Volume overload is a major cause of hypertension in patients receiving PD, particularly when ultrafiltration capacity is insufficient to maintain normotension after RKF declines. Therefore, it is of utmost importance to monitor urine volumes over time and to adjust PD prescriptions when needed to compensate for loss of RKF. Patients with higher peritoneal membrane transport rates should be evaluated for cyclical therapy and/or icodextrin use during the long dwell period, to reduce the risk of volume overload due to long dwell times.

Glucose-containing PD solutions may contribute to cardiovascular disease by increasing a patients' exposure to GDPs (and hence, circulating AGEs) and by possibly inducing insulin resistance, which is associated with cardiovascular morbidity.^{305,306} Thiazolidinediones can be used to modulate insulin resistance, but published evidence in favor of their use is scarce.³⁰⁷ Exercise including strength training may improve glucose tolerance in ESKD with impaired glucose tolerance.^{308,309} Nonglucose PD solutions are another potential solution to this problem; lower serum insulin levels and increased insulin sensitivity have been reported with icodextrin.^{133,310,311} The role of glucose sparing regimens is discussed above.

Hyperlipidemia due to systemic glucose absorption and peritoneal protein losses potentially contribute to the observation of higher levels of total and LDL cholesterol, apolipoprotein B, lipoprotein (a), and triglycerides as well as lower

CARDIOVASCULAR DISEASE IN PERITONEAL DIALYSIS

Cardiovascular disease—including new-onset congestive heart failure, peripheral vascular disease, ischemic heart disease,

levels of high-density lipoprotein cholesterol in patients on PD than in patients on hemodialysis.³¹² As in the general population, statins safely decrease serum cholesterol concentrations in patients receiving dialysis but their efficacy in decreasing cardiovascular mortality has not yet been proven.³¹³ The Study of Heart and Renal Protection (SHARP) included patients who were receiving PD at the time of enrollment or who had imitated therapy with the modality during the course of the trial.³¹⁴ The study demonstrated a lower risk for cardiovascular events in patients treated with simvastatin-ezetimibe, without a beneficial effect on cardiovascular mortality.³¹⁴ Further studies should provide more insight about the efficacy of statins in reducing mortality among patients receiving dialysis, particularly among those on PD, for whom virtually no information is available.

Vascular calcification, affecting the arterial media, atherosclerotic plaques, myocardium, and heart valves, is a common feature in patients with ESKD and is associated with cardiovascular disease and mortality in patients on dialysis, including those treated with PD.³¹⁵ Moreover, cardiac valve calcifications are more frequent in patients on PD who have inflammation, and are associated with a 6-fold higher risk of cardiovascular death.³¹⁶ In patients on hemodialysis, therapy with sevelamer, a non-calcium phosphate binder, attenuates the progression of vascular calcification scores relative to calcium-based binders; effects on survival are uncertain.³¹⁷ Newer, non-calcium binders are now available and future studies with these agents may help elucidate the matter further.

Genetic factors may also affect the risk of vascular complications and outcome in the PD population. Examples are (1) the finding that a single nucleotide polymorphism in the IL-6 gene is associated with higher plasma levels of IL-6, higher diastolic blood pressure, and higher left ventricular mass and (2) the finding that non-BB allele variants of the BsmI polymorphism of the vitamin D receptor gene are associated with increased risk of hypercalcemia.^{318,319} The clinical relevance of these associations, however, remain uncertain.

PERITONITIS

DEFINITION, DIAGNOSIS, AND CLINICAL COURSE

The presence of a cloudy peritoneal effluent is often the earliest sign of peritonitis, although abdominal pain precedes it in some patients. A diagnosis of peritonitis can be established when at least two of the following manifestations are present: abdominal pain, positive result of a peritoneal fluid culture, and more than 100 white cells/mm³ in the dialysis effluent.^{320,321} A bacterial peritonitis typically presents with more than 50% of its WBC as polymononuclear cells. Although turbid effluent is not always associated with peritonitis, its presence should always be considered a sign of possible infection until proven otherwise. When the peritoneal fluid culture is negative and the effluent leukocyte count is normal, other causes for the cloudy effluent should be considered, including the following: fibrin, chylous ascites, malignancy, chemical or eosinophilic peritonitis, and a specimen taken from a dry peritoneum.^{321–326} The presence of more than 10% eosinophils in a cloudy effluent is diagnostic of eosino-

philic peritonitis. The condition may manifest early after catheter placement, or it may be associated with mycotic infections, allergic reactions, and exposure to drugs such as vancomycin.^{327,328}

The most frequent pathway for invasion of the peritoneal cavity by infectious agents is through the catheter's lumen; the next most frequent pathway is the pericatheter route.³²⁹ Transmural bacterial migration has been implicated as another cause of PD-related peritonitis in patients with constipation, diarrhea, recent colonoscopy, diverticulitis, or other bowel pathology. Carriage of *Staphylococcus aureus* in the nasal mucosa and skin is linked to exit-site and catheter-related infections, both of which are risk factors for peritonitis.³³⁰ When purulent exudate is present at the exit site, a swab culture should be performed. Other less common routes of peritonitis are hematogenous spread, either due to active bacteremia or transient bacteremia after periodontal procedures and ascending bacterial spread from the female genital tract. The incubation period of a peritonitis episode is variable depending on the route of infection and the specific organism; for example, touch contamination has been reported to cause symptoms between 6 and 48 hours later.³³¹ The severity of clinical manifestations is also variable between episodes of peritonitis and may be related to the specific causal agent. *Staphylococcus epidermidis* infections in general cause milder symptoms than gram-negative organisms, *S. aureus*, and mycotic organisms.^{320,332} In comparison to surgical or spontaneous peritonitis, PD-related peritonitis is rarely associated with bacteremia.

Bacterial peritonitis typically presents with a peritoneal effluent leukocyte count exceeding 100/mm³ with a leukocyte predominance (effluent neutrophils comprise >50% of the total leukocyte count). Adequate cultures of peritoneal effluent samples should yield positive results in more than 80% of cases. If a high rate of culture-negative peritonitis is present, peritoneal culture methods should be reviewed and optimized. The simplest way to obtain peritoneal fluid cultures is by inoculating 5 to 10 mL of effluent directly into a blood culture bottle; the sample should be sent to the lab within 6 hours or less for best results. Another culture method involves inoculating 50 mL of centrifuged dialysate into a solid culture media; conflicting reports are unable to confirm which method leads to lower rates of culture-negative peritonitis.^{333–335}

Gram stains of PD effluent are notoriously insensitive methods of detecting bacteria, and are useful mainly for early identification of yeast (see [fungal peritonitis](#), below).³³¹ Inadequate effluent collection can cause low leukocyte counts despite clinical symptoms of peritonitis; this occurrence is more common in APD due to short dwell times. If indicative symptoms are present without effluent leukocytosis, cultures and cell counts should be repeated after a dwell period of at least 2 hours. The differential diagnosis of culture-negative peritonitis with dialysate leukocytosis includes inadequate sample collection or laboratory processing, previous antibiotic treatment, or the presence of fungi or mycobacteria.³²⁰

In spite of double-bag systems substituting spike connection in the PD exchange procedure, the most prevalent organism in most reported series is still gram-positive bacteria, particularly *S. epidermidis* acquired through touch contamination. Pathogenic flora may vary significantly according to geographic and other environmental circumstances.^{329,336,337} The second

most frequent bacterial cause of PD-peritonitis is *S. aureus*, often associated with tunnel or exit-site infection; compared with *S. epidermidis*, *S. aureus* peritonitis may cause more severe disease and more often lead to catheter loss and the need to transfer temporarily or permanently to hemodialysis.^{338,339} *Enterococci* are the third most commonly encountered agents of gram-positive peritonitis; they typically respond to antimicrobial treatment but care must be taken to review bacterial sensitivity patterns because vancomycin-resistant strains are not uncommon.³⁴⁰ Constipation and other bowel pathology may predispose to *Enterococcal* infections, which frequently cause recurrent and relapsing peritonitis.³⁴¹

Although gram-positive organisms are the most common cause of PD-peritonitis, the proportion (but not total number) of infections due to gram-negative organisms has increased over the past 20 years.^{342,343} A variety of organisms derived from bowel, skin, urinary tract, water sources, and animal contact have been reported to cause gram-negative peritonitis. Peritonitis from *Escherichia coli* strains and other *Enterobacteriaceae* such as *Klebsiella* or *Proteus* organisms pose serious problems particularly in developing nations, but in most cases, they respond to antibiotic regimens. *Pseudomonas* peritonitis accounts for approximately 8% to 10% of gram-negative peritonitis cases and can frequently necessitate catheter removal. Early catheter removal, temporary transfer to hemodialysis with peritoneal cavity rest, and dual-antipseudomonal antibiotic coverage are all associated with better outcomes, oftentimes allowing a patient to return to PD therapy. *Pseudomonas* species are also commonly associated with exit-site and tunnel infection.^{331,344–346}

Multiorganism gram-negative peritonitis should prompt consideration of a primary abdominal pathologic process, such as diverticulitis or an intraabdominal abscess; however, most patients with polymicrobial peritonitis do not have an underlying abdominal catastrophe.³⁴⁷ In cases of bowel perforation, fecal effluent may be evident and anaerobes are often present. The mortality rate in cases secondary to an abdominal catastrophe may be as high as 50% and is correlated with the primary event, as well as with delayed diagnosis and surgery.³⁴⁸

Fungal peritonitis occurs rarely but is always feared because of high mortality rates; early peritoneal catheter removal is required.³⁴⁹ The species most frequently involved is *Candida albicans*; the concurrent presence of bowel obstruction and abdominal pain portend worse outcomes. It has been postulated that current or recent antibacterial treatment may increase the risk for fungal peritonitis; systemic antibiotics (for peritonitis or even non-PD infections) are thought to suppress normal bowel flora, thus allowing intestinal overgrowth of fungi; fungal peritonitis then can result due to fungal mural transmigration from the intestine into the peritoneum.³⁵⁰

Peritonitis caused by mycobacteria is very rare; in most instances, it is initially diagnosed as culture-negative peritonitis. Diagnosis requires adequate culture techniques and a high index of suspicion. In mycobacterial infections, leukocyte counts generally exhibit a predominance of polymorphonuclear leukocytes in the peritoneal effluent.^{320,351} Acid-fast smears are notoriously insensitive and mycobacterial cultures require days to weeks of incubation time. Polymerase chain reaction techniques have been useful in detecting tuberculous gene products in peritoneal fluid, and there are

reports of retrospective diagnosis either after peritoneal biopsy (at the time of catheter removal) or after empiric anti-mycobacterial therapy led to a clinical resolution of the infection. In areas that are endemic for tuberculosis, tuberculosis peritonitis may comprise up to 4% of all infections and carries a high morbidity, particularly in patients with protein-energy wasting.^{352,353}

PERITONITIS TREATMENT

Empirical antibiotic treatment should be started as soon as infectious peritonitis is diagnosed, i.e., within hours. Peritoneal fluid leukocytosis is sufficient cause to initiate antimicrobial therapy, and treatment should start immediately without delay even if culture results are pending.³²⁰ Initial empiric antibiotic regimens should include both gram-negative and gram-positive coverage. Thereafter, antibiotic regimens should be tailored to culture and sensitivity reports once available; center-specific bacterial sensitivity patterns should be taken into consideration when choosing antimicrobial therapy. Cloudy effluent and abdominal pain usually improve early in the course of treatment and may disappear within 48 to 72 hours. Persistent clinical manifestations of peritonitis may indicate a nonresponding bacterial organism, and necessitates prompt change in antibiotic treatment or consideration of catheter removal.³²⁰ Follow-up effluent cell counts, preferably within the first 5 days, are a useful tool to evaluate treatment response because prolonged peritoneal fluid leukocytosis is associated with a higher rate of treatment failure and need for catheter removal.^{354–356}

INITIAL EMPIRIC THERAPY

Ideal empirical peritonitis treatment should provide broad antimicrobial coverage for the most frequent causes, should be convenient to administer, and should not favor the development of resistant organisms.³⁵⁷ Such a perfect regimen does not exist. Selection of empirical therapy should be center-specific, depending on the local history of sensitivities of organisms. Empirical antibiotics must cover both gram-positive and gram-negative organisms. The latest international guidelines, based on evidence when available, include a regimen containing a cephalosporin or vancomycin for gram-positive bacteria together with either a third-generation cephalosporin, such as ceftazidime or cefepime, or an aminoglycoside.³²⁰ Vancomycin is highly effective against most gram-positive bacteria, but its use has raised discussions about whether it favors the appearance of vancomycin-resistant staphylococci and enterococci.³⁵⁸ Drug-resistant *S. aureus* and *Enterococcus* strains may necessitate administration of newer agents such as linezolid, quinupristin/dalfopristin, carbapenems, or daptomycin.³⁵⁹

Changing microbiologic features, toxic effects of drugs, or difficulties administering therapy may lead individual clinical groups to tailor their initial antimicrobial regimen to their own patients' needs. Evidence exists from several prospective studies that monotherapy with different agents (aztreonam, oral quinolones, cefepime) may be efficacious, but results are controversial.³⁵⁹

Antibiotic dosages should be determined based on the patient weight and RKF; when not adjusted for patient weight and RKF there is a considerable risk for underdosing with the result of an increased risk of treatment failure.

ROUTE AND SCHEDULE OF ADMINISTRATION

Antibiotics can be administered by different routes (e.g., oral, intraperitoneal, intravenous). Intraperitoneal (IP) administration is much preferred to intravenous infusion; IP administration produces a greater concentration of antibiotics locally at the infection site, it is easy to administer, and has been demonstrated to be superior in a meta-analysis of randomized controlled trials (Table 64.8).³⁵⁹ According to this analysis, intermittent (once a day in a long dwell of at least 6 hours) and continuous (antibiotics with each exchange) administration of antibiotics are equally efficacious in CAPD, and there is no benefit for routine peritoneal lavage or use of urokinase.³⁶⁰ Antibiotic dosing and interval of administration in APD has not been sufficiently studied for all available regimens. The most recent guidelines acknowledge the lack of evidence for adequate antibiotic prescription in APD and guidelines are available for intermittent dosing of antibiotics for such patients (Table 64.9); nevertheless, they propose continuous dosing particularly of cephalosporins for this modality.³²⁰ Cephalosporins and aminoglycosides could be administered intermittently in the long-dwell daytime exchange.^{361,362} Vancomycin is administered intermittently every 3 to 5 days based upon the patient's own drug metabolism, and blood levels may be monitored to guide therapy, at least initially. Since vancomycin efficacy is time-dependent, care should be taken to maintain therapeutic blood levels above the bacteria's minimum inhibitory concentration (MIC) for as long as possible during the entire dosing interval. Researchers recommend keeping the peak value 5 to 8 times the MIC and that the trough value should be 1 to 2 times the MIC in order to reduce the risk for the development of resistance.³⁶³

SPECIFIC ANTIBIOTIC TREATMENT

Once the results of the peritoneal effluent culture are known, if the initial empirical treatment is inadequate, it should be adjusted and the response monitored.

Gram-Positive Bacteria

Coagulase-negative staphylococci usually respond rapidly to treatment and are adequately eradicated with the initially prescribed cefazolin or vancomycin, subsequently administered for 2 weeks.³²⁰ Treatment of *S. aureus* peritonitis can continue with a first-generation cephalosporin if the organism is methicillin sensitive, or with vancomycin if it is methicillin resistant. In the rare occurrence of vancomycin-resistant *S. aureus*, linezolid, daptomycin, or quinupristin/dalfopristin should be prescribed. *S. aureus* infections should be treated for 3 weeks, even if a clinical response is seen early in the infection course. In the presence of an exit-site infection by the same microorganism, the catheter often must be removed.^{320,338–340} If enterococcal peritonitis is diagnosed, intraperitoneal ampicillin and an aminoglycoside are indicated, or vancomycin in the case of ampicillin-resistant enterococci for 3 weeks.³²⁰

Gram-Negative Bacteria

Non-pseudomonal gram-negative enterobacteria usually respond to third-generation cephalosporins or aminoglycosides; dual-antibiotic coverage may reduce the relapse and recurrence rate of certain gram-negative peritoneal

Table 64.8 Intraperitoneal Antibiotic Dosing Recommendations for Patients on CAPD^a

Drug	Intermittent (Per Exchange, Once Daily)	Continuous (mg/L ^b ; All Exchanges)
Aminoglycosides		
Amikacin	2 mg/kg	LD 25, MD 12
Gentamicin, netilmicin, or tobramycin	0.6 mg/kg	LD 8, MD 4
Cephalosporins		
Cefazolin, cephalothin, or cephadrine	15 mg/kg	LD 500, MD 125
Cefepime	1000 mg	LD 500, MD 125
Ceftazidime	1000–1500 mg	LD 500, MD 125
Ceftizoxime	1000 mg	LD 250, MD 125
Penicillins		
Amoxicillin	ND	LD 250–500, MD 50
Ampicillin, oxacillin, or nafcillin	ND	MD 125
Azlocillin	ND	LD 500, MD 250
Penicillin G	ND	LD 50,000 U, MD 25,000 U
Quinolones		
Ciprofloxacin	ND	LD 50, MD 25
Others		
Aztreonam	ND	LD 1000, MD 250
Daptomycin ¹¹⁵	ND	LD 200, MD 20
Linezolid ⁴¹	Oral 200–300 mg qd	
Teicoplanin	15 mg/kg	LD 400, MD 20
Vancomycin	15–30 mg/kg every 5–7 days	LD 1000, MD 25
Antifungals		
Amphotericin	NA	1.5
Fluconazole	200 mg IP every 24–48 hours	
Combinations		
Ampicillin/sulbactam	2 g every 12 hours	LD 1000, MD 100
Imipenem/cilastatin	1 g bid	LD 250, MD 50
Quinupristin/dalfopristin	25 mg/L in alternate bags ^c	
Trimethoprim/sulfamethoxazole	Oral 960 mg bid	

^aFor dosing of drugs with renal clearance in patients with residual kidney function (defined as >100 mL/day urine output), dosage should be empirically increased by 25%.

^bExcept as noted.

^cGiven in conjunction with 500 mg intravenous twice daily.

From Li PK-T, Szeto CC, Piraino B, et al: Peritoneal dialysis-related infections recommendations: 2010 update. *Perit Dial Int* 30:393–423, 2010.

Table 64.9 Intermittent Administration of Antibiotics in Automated Peritoneal Dialysis

Drug	Intraperitoneal Dosage
Cefazolin	20 mg/kg every day, in long daytime dwell ¹¹⁴
Cefepime	1 g in one exchange per day
Fluconazole	200 mg in one exchange per day every 24-48 hours
Tobramycin	LD, 1.5 mg/kg in long dwell; then 0.5 mg/kg each day in long dwell ¹¹⁴
Vancomycin	LD, 30 mg/kg in long dwell; repeat 15 mg/kg in long dwell every 3-5 days (aim to keep serum trough levels above 15 µg/mL)

infections.^{344,346} *Pseudomonas aeruginosa* peritonitis should be treated with two antibiotics for three weeks; if clinical improvement is not seen promptly, catheter removal is indicated. In individuals with coexisting exit-site infection with the same organism, catheter removal is recommended because of a high incidence of relapse even if there is clinical improvement with antibiotics alone.³²⁰ Intraperitoneal aminoglycosides and third-generation cephalosporins, oral quinolones, and intravenous piperacillin have been employed with success in pseudomonal peritonitis.³⁴⁸ Aminoglycosides should be used with caution; in addition to effects on RKF, aminoglycosides can result in serious vestibulotoxicity. Given the lower efficiency of solute clearance in PD, higher time-averaged exposure to aminoglycosides may be seen, placing patients at higher risk of adverse effects.

Polymicrobial Peritonitis

In the presence of multiple enteric organisms, catastrophic intraabdominal pathologic processes such as diverticulitis or appendicitis should be considered and, if diagnosed, surgery performed.³⁴⁸ Antibiotic treatment should include an aminoglycoside, a third-generation cephalosporin or carbapenem, and anaerobic coverage with metronidazole or clindamycin. Treatment should continue for at least 3 weeks.

Fungal Peritonitis

Fungal peritonitis is a potentially lethal infection that responds poorly to antifungal agents if the catheter is not removed because the organism forms biofilm on the catheter, thus resulting in a permanent reservoir for the fungus.^{349,350} Occasionally, patients with fungal peritonitis may be too ill for surgery or may refuse catheter removal. Small case reports describe successful treatment of fungal peritonitis without catheter removal by using a combination of systemic and IP antifungal therapy plus continuous high-dose intracatheter amphotericin B.³⁶⁴ With respect to selection of antifungal agents, a combination of flucytosine and amphotericin B has traditionally been recommended; development of resistance is high with flucytosine so this agent should never be used alone. Newer, less toxic antifungal agents (e.g., voriconazole, fluconazole, or caspofungin) may be successful when chosen according to fungal culture results and antifungal sensitivity reports; infectious disease consultation can be helpful in choosing proper antifungal therapy in complicated situations.^{320,350} Treatment typically continues for 2 to

4 weeks once the catheter is removed.³²⁰ There is a high mortality rate associated with keeping the catheter in situ; therefore fungal peritonitis should always require catheter removal.³⁴⁹ Intraperitoneal abscess and adhesion formation are the most common reasons patients cannot return to PD after microbiologic cure.³⁶⁵ Some have advocated keeping the catheter in place for a short period of time in order to administer IP antifungals, in an attempt to reduce the chance of adhesion formation.

Mycobacterial Peritonitis

Treatment of mycobacterial peritonitis requires a complex antibiotic regimen including isoniazid, pyrazinamide, ofloxacin, and intraperitoneal rifampicin. Catheter removal may be required and treatment with four agents should commence as soon as possible, but definitely within 4 to 6 weeks of initial presentation to produce the best outcomes. Treatment should continue for 6 to 9 months, incorporating an initial intense treatment with four drugs followed by a maintenance period with two antimycobacterial agents.^{351,352}

CATHETER REMOVAL IN PERITONITIS

In addition to fungal peritonitis, intraabdominal disease, and refractory tunnel and/or exit-site infections, two other conditions warrant catheter removal: (1) relapsing peritonitis, defined as an episode with the same organism that caused the preceding episode of peritonitis, or one sterile peritonitis within 4 weeks after the end of the initial course of antibiotics, or (2) refractory peritonitis, defined as the failure to respond within 5 days of appropriate antibiotics. Catheter removal and simultaneous placement of a new catheter is most successful for exit-site or tunnel infections, or in relapsing infections once the effluent cell count and future have normalized; it should not be done in refractory peritonitis or in severe peritoneal infections³⁶⁶; success is most common in infections that do not involve *S. aureus*, *Pseudomonas*, mycobacteria, or fungi – in these cases the patient should be placed on hemodialysis until peritonitis is resolved (usually 3 to 4 weeks) before the placement of a new catheter.³²⁰

PERITONITIS PREVENTION

Adequate catheter placement is relevant for peritonitis prevention. Exit-site placement and prophylactic antibiotic therapy at the time of placement, usually with a first-generation cephalosporin, is used to prevent postoperative infections. Only when the exit site is healed should the patient take over care of the exit site. Mupirocin cream applied to the exit site has proved useful in preventing local *S. aureus* infection. Mupirocin applied to the nares 5 days a month is also effective in reducing *S. aureus* exit-site infection, however it is ill-tolerated and requires monthly therapy. The use of local gentamicin has been shown to reduce *Pseudomonas* and other gram-negative infections at the exit site, as well as peritonitis rates.³⁶⁷ The training of patients should include teaching them how to identify and notify the dialysis facility that contamination has occurred; episodes of contamination can be managed with change of transfer set with/without administration of prophylactic antibiotics depending upon whether a minor or major contamination occurred.³⁶⁸ The use of double-bag systems in ambulatory patients and a

flush-before-fill step after connection of the tubing to the solution bags in automated procedures are highly efficacious in preventing peritonitis episodes.^{368–372}

NONINFECTIOUS COMPLICATIONS OF PERITONEAL DIALYSIS

MECHANICAL COMPLICATIONS

Intraperitoneal instillation of dialysate causes an increase in intraabdominal pressure, and the amount of intraabdominal pressure varies depending upon several factors including age, body mass index, volume of dialysate, and patient position, where intraabdominal pressure is highest in the sitting position, less when standing, and least in the supine position.^{373,374} Additionally, certain maneuvers such as coughing, straining during defecation, and lifting may further increase intraabdominal pressure. The major risks associated with increased intraperitoneal pressure are the development of hernia, pericatheter leaks, diaphragmatic leaks, restriction of pulmonary expansion with resultant dyspnea, gastroesophageal reflux, abdominal discomfort, and pain.³⁷⁵

Hernias occur in more than 10% of patients on peritoneal dialysis. Many hernias, particularly in the inguinal or periumbilical region, are present prior to start of PD and become more apparent with the intraperitoneal instillation of dialysate. Hernias manifest as several different types including umbilical, abdominal (ventral), incisional, and indirect inguinal hernia. A preexisting but previously undetected patent processus vaginalis can suddenly become apparent as an inguinal hernia after initiation of PD due to new dialysate flow into the hernia, causing genital swelling.³⁷⁶ New hernia can form *de novo* at the catheter incision site, umbilicus, ventral abdominal wall, or inguinal areas. Most hernia require surgical repair; however, conservative treatment may be indicated for some, particularly in elderly patients. Surgical repair may be performed successfully without temporary transfer to hemodialysis if the patient can be treated with low-volume supine exchanges for 2 to 4 weeks postoperatively.³⁷⁷ For cases requiring mesh placement, surgical techniques have been described wherein the mesh can be placed extraperitoneally.³⁷⁸ Ultimately, however, the need for surgery must be judged and decided on an individual basis.^{376–378}

Pericatheter leaks may occur after dialysis initiation and are more common in obese patients. They may manifest as an external dialysate leak or as abdominal or genital edema. Conservative treatment, often successful, includes reduction of inflow volume per exchange, nocturnal intermittent PD, or temporary PD postponement and, if needed, temporary transfer to hemodialysis.^{375,376,378–382} If leakage recurs, the catheter may require reinsertion.^{378,348} A 3- to 4-week break-in period, defined as the time between catheter insertion and initial use, significantly reduces the risk of pericatheter leaks.^{375,378}

Diaphragmatic leaks, due to preexisting diaphragmatic stoma, are uncommon and become typically clinically apparent soon after initiation of PD. The diagnosis of pleuro-peritoneal fistula may be made using imaging techniques whereby contrast dye or radioactive isotope is instilled into the dialysate solution, and later found in the pleural space. Upright daytime-only dialysis (with an empty peritoneum at night)

can be performed for very temporary relief, and ultimately surgical correction or pleurodesis is needed to continue PD.

METABOLIC COMPLICATIONS

The most relevant metabolic complications associated with PD are related to the consequences of systemic glucose absorption. Absorption of glucose from PD solutions may provide 500 to 800 kcal/day.^{382–386} Glucose absorption may induce hyperglycemia in previously nondiabetic PD patients with impaired glucose tolerance.³⁸⁶ Treatment of hyperglycemia may be a complex clinical problem. Biguanides are contraindicated, as in all patients with ESKD, but other hypoglycemic agents, including some sulfonylureas and thiazolidinediones, may be used.³⁸⁴ The latter have been associated with reduction of fibrosis by inhibiting inflammation and regulating the transforming growth factor/SMAD signaling pathway.³⁸⁷ Intraperitoneal insulin has been used in diabetic patients with CAPD because it appears convenient and physiologically beneficial, but it poses an important risk of contamination in the process of bag injection. Use of non-dextrose containing PD fluids has been associated with improved glycemic control.

A second metabolic condition highly prevalent in patients on PD is the presence of hyperlipidemia, accompanied by high levels of LDL cholesterol and apolipoprotein B; this lipid profile may be atherogenic and potentially contribute to the high cardiovascular mortality among patients on PD, although direct evidence of a link is lacking.³⁸⁸ Patients may be treated with fibrates if hypertriglyceridemia is severe; statins are also used to lower levels of low-density lipoprotein cholesterol.³⁸⁹ Combined use of these two agents is not recommended because of risk for rhabdomyolysis and/or hepatotoxicity.

Abnormalities in mineral metabolism are frequently present in patients on PD. Hypercalcemia may occur in PD due to vitamin D use, high calcium levels in some peritoneal fluids (i.e., 3.5 mEq/L), and ingestion of calcium-based phosphate binders; this in turn may induce suppression of parathyroid hormone secretion resulting in adynamic bone disease, a condition which has been reported to occur more frequently in PD than in hemodialysis.^{367,390,391} The increased use of lower calcium dialysate (i.e., 2.5 mEq/L) and the use of non-calcium phosphate binders has reduced the incidence of hypercalcemia with PD. Phosphate control tends to be better in PD than in hemodialysis, probably from better adherence to diet and/or binders in patients who have additionally assumed the responsibility for their own dialysis, the continuous nature of the dialytic procedure, and longer maintenance of RKF.^{390–392} Nevertheless, similar associations of hyperphosphatemia and mortality have been observed among patients on PD and HD.³⁹³

Among all patients receiving dialysis, those on PD are far more likely than those on HD to experience hypokalemia. Hypokalemia is not uncommon in PD, owing to the continuous nature of PD and the use of dialysate devoid of potassium; additionally, low potassium intake and/or high dose of loop diuretic therapy can also contribute to hypokalemia. Fortunately, nearly all patients on PD can achieve normal serum potassium concentrations by increasing dietary intake of potassium-containing foods and/or oral potassium supplements. Hyperkalemia is less commonly encountered in PD

than hypokalemia and is usually related to renin-angiotensin system blockade, missed dialysis or dietary excess. It is usually self-limited and not severe, as long as the patient is adherent with the dialysis prescription.

Hyponatremia is a common finding in patients on PD, partly because of fluid overload and the low dialysate sodium concentration, usually 132 mmol/L. Severe hyponatremia is infrequent and may be associated with hyperglycemia, protein-energy wasting, or water overload.^{394,395} Hypernatremia is very rarely observed but may be present in older patients who have lost the sense of thirst or have limited access to water; a long series of frequent exchanges with very short dwell periods using hypertonic dialysate can also lead to hypernatremia through the mechanism of sodium-sieving, discussed above. Sodium sieving is easily avoided by allowing some longer dwell periods throughout the 24-hour cycle, thus allowing sodium to re-equilibrate between blood and dialysate.

ENCAPSULATING PERITONEAL SCLEROSIS

Encapsulating peritoneal sclerosis (EPS), one of the most feared complications of PD, is an uncommon but serious entity that is most often associated with longer time on PD; the cumulative incidence varies between 0.5% to 4.4%.^{396,397} In this condition, massive sclerosis of the peritoneal membrane ensues, resulting in encapsulation of the intestines. Encapsulating peritoneal sclerosis is responsible for severe disturbances of intestinal function, manifesting as motility disorders that cause impaired nutrient absorption, obstructive ileus, hemorrhagic ascites, anorexia, weight loss, and progressive clinical deterioration. Systemic inflammation is usually present, manifested by low-grade fever, hypoalbuminemia, elevated levels of serum C-reactive protein, and other inflammatory markers. The diagnosis requires clinical features of intestinal obstruction or disturbed gastrointestinal function and evidence of bowel encapsulation, either radiologically or pathologically. Computed tomography (CT) scanning is a reliable tool and can be used to confirm a diagnosis of EPS in an appropriate clinical setting.^{398–402} Pathologic confirmation is available when patients are subjected to surgery for treatment or catheter removal.^{403–405} However, care should be exercised because accidental injury to bowel may result in the formation of entero-cutaneous fistula.

The cause of encapsulating peritoneal sclerosis is unknown, but a number of factors that may contribute or predispose to its development have been identified. Contributing factors can be divided into those directly related to PD (time on PD, peritonitis, plasticizers, bio-incompatible dialysate, discontinuation of PD) and those that are not (idiopathic, β -blockers, autoimmune diseases, cancer, talc or other particulate substances, genetic predisposition).^{403,404} Unknown factors render some patients more susceptible, inasmuch as some never develop encapsulating peritoneal sclerosis even after continuous exposure to multiple predisposing conditions. The reported incidence is higher in some countries, particularly Japan and Australia and an apparent increase in incidence has been reported in the European Union.^{403,404,406} This increase may be related to ethnic or genetic factors, increased longevity of PD technique in certain areas, longer waiting times for kidney transplantation, or due to improved diagnosis and awareness of the disease. Among patients with a confirmed

diagnosis, the mortality rate is very high, varying from 20% to more than 90%.^{404,406} However, the survival in recent cohorts appears considerably higher than the earlier reports,⁴⁵⁷ possibly related to a greater recognition of the disease entity and diagnosis at an early stage than in previous years.

Treatment for encapsulating peritoneal sclerosis is often unsuccessful and could even be described as ineffective, particularly if it is not implemented early in the course of the disease. There is no defined treatment of choice. Surgical treatment involves releasing or lysing adhesions of the small bowel, and requires precision and expertise to avoid morbid outcomes such as enterocutaneous fistula.⁴⁰⁷ Medical treatments are either supportive or therapeutic, the latter aimed at ameliorating the inflammatory and pro-fibrotic processes in EPS. Total parenteral nutrition and discontinuation of PD with subsequent transfer to hemodialysis are often required; in some (but not all) instances, such treatment may induce regression of the pathogenic process.^{403,404,406,408,409} One promising therapeutic option in EPS is the antifibrotic agent tamoxifen.^{410,411} Prevention of encapsulating peritoneal sclerosis is not yet possible because clinicians do not understand the process that leads to this condition; nevertheless, early identification may be of great importance because early intervention is more effective than treatment later in the disease process.^{403,406,408,409}

PATIENT OUTCOMES WITH PERITONEAL DIALYSIS

Optimal comparison of two therapies, such as HD and PD, requires adequately powered randomized controlled clinical trials. One such trial had been attempted in the Netherlands but had to be abandoned as over 90% of eligible patients refused to be randomized.⁴¹² A more recent clinical trial in China has been closed for recruitment without achieving its target enrollment and results are awaited ([clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT01413074) identifier: NCT01413074). Up until the results of this trial is available, the information available is based on national registries and a few prospective cohorts from around the world.^{413–424}

Results of these studies are not conclusive and discrepancies can be attributed to multiple factors, including methodological issues such as use of intent-to-treat versus as-treated analysis, degree of case-mix adjustments, use of proportional versus nonproportional hazards models, assessment of prevalent versus incident patients, among others.^{425–427} In general, when these differences are accounted for, similar results are found among registry studies and, to a lesser degree, among prospective cohort studies.^{420,422,423}

Nondiabetic and younger patients treated with PD have an equal or lower risk for death compared to those treated with HD; among older diabetic patients, results vary by country.⁴²⁵ For example, the Canadian registry showed no difference in risk for death between PD and HD among older diabetic patients and similar results were reported in Korea,⁴²⁸ whereas in the United States, older diabetic patients treated with HD had a lower risk for death than those treated with PD.⁴²⁰ In Danish and Norwegian populations, recent studies demonstrate a similar or lower risk for patients undergoing PD, which could be related to a cohort effect and to the mode of dialysis initiation.^{429,430} Another Korean

study demonstrated that patients with cardiovascular disease or diabetes on PD had a lower survival than those undergoing HD.⁴³¹ A recently published analysis of the REIN Registry demonstrated that patients with heart failure treated with HD had a lower risk for death than the patients treated with PD.⁴³² Similarly, the study of Wang and coworkers demonstrated that PD was associated with poorer survival among patients with cardiovascular disease or diabetes mellitus.⁴³¹ Despite disparate findings, it remains unclear whether the differences in risk for death are a result of differences in patients treated with the two modalities, or a direct benefit (or harm) of the therapy itself.^{426,427} Hence, at present, these studies should inform but not be central to the selection of dialysis modality for an individual patient.⁴³³

Multiple studies have demonstrated a time-dependent trend in the relative risk of death, whereby PD is generally associated with equivalent or better survival rates during the first 1 or 2 years of dialysis. A European study reported that patients on PD had better survival during the first year and HD patients had better survival in the next 2 years of dialysis, yet overall survival comparing HD and PD was similar and influenced by age, diabetes mellitus, and center size.⁴³⁴ Recent studies suggest that some of these differences are because of bias: many patients who start treatment with HD have had inadequate preparation for dialysis, which in turn, is an independent risk factor for death. When accounting for this source of bias, there does not seem to be significant differences in early survival among patients treated with PD or HD.^{420,435}

When long-term survival is analyzed, results vary according to the nature of the study and to different subgroups within studies. Although subgroup analyses in the cohort studies have the advantages of being prospective and provides more clinical and laboratory details, they are usually limited by smaller numbers of patients in comparison with larger registry-based studies.^{417,436,437} Large registry-based studies, although offering adequate statistical power for subgroup analyses, may be “overpowered” for detecting differences between the whole population of subjects on PD and those on HD. For instance, the overall relative risk of death in PD versus HD was 1.04 (95% confidence interval = 1.03 to 1.06, $P < 0.001$) for U.S. Medicare recipients who initiated dialysis between 1995 and 2000⁴¹⁴; this difference was statistically significant in being powered by nearly 400,000 patients studied. However, this relative risk of 1.04 translates into an adjusted 3-year survival difference between HD and PD of 1 month.⁴²⁵ A general improvement in survival among patients with renal replacement therapy has been observed; these improvements seem to be greater for patients undergoing PD than those treated with HD.^{419,438,439} According to the 2013 United States Renal Data System (USRDS) Annual Data Report, there is a continued trend to an improvement in survival in patients who begin either HD or PD as compared to data between 1993 and 1997 as well as data from 1998 and 2002.⁴⁴⁰ Data from other countries of the world, in particular Europe and Japan, have demonstrated better survival than data from the USRDS, results initially attributed to higher quality of treatment yet multiple factors including patients’ comorbidities, age, and other factors undoubtedly play an important role in these differences, which make comparisons quite difficult.

As noted above, most studies demonstrate similar survival between PD and conventional HD. Therefore, modality choice

must be individualized and tailored to maximize patient-reported outcomes including health-related quality of life.⁴⁴¹

Patients on PD experience technique failure, most often due to peritonitis, catheter malfunction, “burn-out”, and in some long-term patients, ultrafiltration failure; such patients are transferred to HD.^{442,443} This fact should not be regarded as a limitation of PD, as in an appropriate renal replacement therapy program, patients must be requiring to be transferred from one modality (PD, HD, or transplantation) to another according to medical and other needs.

In summary, there is no conclusive evidence that one modality is superior to the other in terms of survival rate for all patients with ESKD, yet some subpopulations seem to perform better than others in PD and others in general do better in HD. It is important to point out that an individualized and educated decision is essential in deciding renal replacement modality and may play a role in survival.^{441,444} Appropriate clinical judgment, consideration of cultural factors and informed patient choice are factors that should be always taken into account in deciding which dialytic modality will be chosen.

ECONOMICS AND COST EFFECTIVENESS OF PERITONEAL DIALYSIS

ESKD is a constantly growing problem worldwide, and its treatment is extremely expensive. According to the 2015 USRDS report, the overall Medicare expenditure for ESKD treatment was \$33.9 billion in the United States.⁴⁴⁵ The economics of dialysis is very complex, and dialysis cost may threaten the health care systems of many countries.⁴⁴⁶ In the industrialized world, PD has been shown to be less expensive than HD.^{447–449} Most of the studies are either limited cost analyses or considerations of costs in relation to survival. Some studies have also shown that the cost-utility ratio is most favorable for PD.⁴⁵⁰ In general, the cost continuum, from most to least expensive, is as follows: in-center HD, out-of-center HD (satellite-, or self-care HD) similar to APD, home HD, and CAPD.

Several reasons may explain the lower cost of PD, including the fact that the patient or a helper administers dialysis, whereas HD must be performed by trained staff, which is relatively expensive. In addition, PD requires fewer physical resources than does HD. Therefore, in countries with predominantly public dialysis providers such as government-run hospitals (Great Britain, Canada, New Zealand, Hong Kong, and Mexico), the use of PD is much higher than in countries with mainly private providers, such as physicians’ offices or dialysis chains (United States, France, Germany, and Japan) in order to constrain ESKD expenditure.

The question that arises next is: If PD is less costly, why do private providers continue to prescribe it in a small percentage of cases? Several factors may explain this phenomenon, but the underlying argument is that the economic drivers of modality selection in countries with mainly private providers are determined by the local economic features of the system and not by actual costs (HD is more profitable).^{451,452} In this situation, it is important to consider the following: (1) Once an HD unit is set up, there is economic pressure to maximize its efficiency by operating it at full capacity; (2) PD may cost less, but the payer (frequently the government) may

reimburse expenses at a correspondingly lower level; and (3) in many countries with mainly private providers, HD offers more opportunity to bill for additional procedures (e.g., provision and administration of intravenous erythropoietin, vitamin D analogs, and iron).⁴⁵¹ An analysis of reimbursement policies and regulations in seven developed nations gives further insight to the complexity and diversity of factors that may influence dialysis modality selection and further stresses the need for better research in this area.⁴⁵³ Changes implemented in 2010 in ESKD reimbursement policies in the United States, in which a bundled payment per person is considered, independent of modality chosen, continues to promise an increase in the use of PD.^{452,454} A factor that significantly reduces the use of PD in some regions is a lack of adequate PD education and training in some nephrology programs, as a result of lack of expertise, interest, or economic incentive to promote PD growth. This has been reported as a significant factor limiting the increase of PD in the United States.⁴⁵⁵

The economics of dialysis in developing nations differs from that of industrialized nations.⁴⁴⁶ In developing nations, where labor is relatively inexpensive and the cost of imported equipment and solutions is high, PD could be more expensive than HD. A further negative aspect for PD therapy in developing nations is that treatment may still entail outdated intermittent systems, and sometimes transfer sets are even reused, with high rates of peritonitis, increased costs, and high dropout rates. Several published analyses disavow this perception, showing that PD treatment may be less expensive than HD in developing nations⁴⁵⁶; in fact, the local production of PD fluids and competition between different providers may influence the lowering of PD costs. Thus, the perception that PD is not a viable option for developing countries is clearly inappropriate. In countries where the cost of PD is lower than that of HD, an increased use of clinically appropriate PD provides an opportunity to substantially lower overall the cost of ESKD treatment, as is being demonstrated in some nations.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. A young adult female with end stage renal disease has been undergoing APD for the last four months. After five months of amenorrhea, she started her menstrual periods yesterday and last night, when she connected to the cycler, noted that the dialysate in the initial drain was pink in color. She reported to the dialysis unit the next day and the PD nurse performed several in-and-out exchanges and but the fluid remained pinkish-red. The PD fluid sample was sent for cell count: 20 WBCs/mm³ and 800 RBCs/mm³ were present.

Which ONE of the following represents the best plan of action in this patient?

- Intravaginal ultrasound to evaluate for ectopic pregnancy
- Antibiotic treatment for presumed peritonitis
- Intraperitoneal heparin for the period of hemoperitoneum
- Peritoneal fluid cytology to exclude peritoneal carcinomatosis

Answer: c

Rationale: Occurrence of hemoperitoneum is not unusual in young, menstruating women at the time of menstrual flow and is the consequence of retrograde movement of blood through the fallopian tubes into the peritoneal cavity. This is a benign and self-limited process. In addition, in this patient, the peritoneal WBC count is less than the threshold of 100/mm³ required for the diagnosis of PD-related peritonitis. The greatest risk in this patient with hemoperitoneum is that the blood will clot in the lumen of the PD catheter and lead to catheter dysfunction. Thus if the PD fluid fails to clear after in-and-out exchanges, addition of heparin to the dialysate to prevent the blood from clotting is recommended.

2. A 50-year-old male patient who is starting on CAPD with twin bag disconnect system is seeking advice to minimize the risk for PD-related peritonitis. Which ONE of the following has been demonstrated to be consistently effective in reducing risk of PD-related peritonitis?

- Use of icodextrin for the long overnight dwell
- Insertion of a pre-sternal peritoneal dialysis catheter
- Use of sterile gloves for performing peritoneal dialysis exchanges
- Daily application of an antibiotic cream of mupirocin or gentamicin at the peritoneal dialysis catheter exit site

Answer: d

Rationale: Daily local application of either mupirocin or gentamicin at the exit site of the catheter has been shown to be effective in reducing peritonitis rates in patients treated with PD. Hand hygiene is critical in reducing infectious risk related to the performance of a PD exchange, nevertheless, it is not necessary to use sterile gloves. The use icodextrin does not modify the risk of peritonitis. Similarly, the insertion of a pre-sternal catheter does not offer any benefit in terms of risk reduction or PD-related peritonitis.

3. With regards to residual renal function in the PD patient, which of the following statements is correct?

- High dose diuretics slow the rate of decline in residual renal clearances in patients undergoing peritoneal dialysis.
- Patients undergoing PD maintain their residual renal function longer than those undergoing HD.
- Renal and peritoneal clearances have an equivalent effect on survival
- Use of N-acetylcysteine at the time of radio-contrast procedures has been shown to reduce the rate of loss of residual function in patients undergoing peritoneal dialysis

Rationale: b

Rationale: It has been clearly demonstrated that patients on PD maintain their renal residual function longer than patients on HD. In addition, CANUSA and other studies have demonstrated that renal and peritoneal clearances are not equivalent for survival. High dose diuretics do not protect from the loss of RRF. The use of N-acetylcysteine has not demonstrated to protect against loss of residual function in PD or other patients.

4. A 54-year-old diabetic man on PD for three years presents at the emergency room referring abdominal pain of 8 hours and the presence of cloudy effluent at dialysate drainage. At examination he presents with normal body temperature, abdominal tenderness, and unremarkable examination of the peritoneal catheter exit site and tunnel. The dialysate was confirmed to be cloudy and a sample of PD effluent was sent for gram stain, cell count, and culture. Preliminary report: 800 WBCs/mm³.

Which of the following options is the optimal approach for empiric treatment of PD-related peritonitis?

- Mono-therapy with ciprofloxacin
- Combination therapy with vancomycin and rifampin
- Mono-therapy with intraperitoneal amikacin
- Combination therapy with cefazolin and ceftazidime

Answer: d

Rationale: Present guidelines suggest the use of two medications for empiric therapy for PD-related peritonitis. One of the antibiotics should provide appropriate coverage gram positive organisms and a second one for gram negative organisms. The combination of cefazolin and ceftazidime meets this requirement (option d). Monotherapy with ciprofloxacin is inappropriate as it covers only some gram positive and gram-negative organisms and its use leads to rapid emergence of resistance. Combination of vancomycin and rifampin would also be inappropriate as it provides gram positive coverage but not for gram negative organisms. Monotherapy with amikacin is avoided as it provides coverage only for gram negative microorganisms and induces the potential risk of loss of renal residual function due to nephrotoxicity.

5. A 60-year-old man with end-stage renal disease secondary to nephrosclerosis is undergoing treatment with automated PD for three years. He is 5'7" tall and weighs 182 lbs (body surface area, 1.93 m²; body mass index, 29 kg/m²). Automated PD prescription: 4 overnight exchanges of 2.5 L and one 2.5 L last bag fill with dextrose-based solution. Over the last year he has been progressively losing his renal residual function until present time in which he is anuric. The weekly Kt/V_{urea} on this last 24-hour dialysate collection, completed last week was 1.56.

Which ONE of the following is the most effective initial approach to increase the small solute clearances in this patient?

- a. Add a 2.5 L day-exchange following the last-bag-fill
- b. Transfer to hemodialysis
- c. Change the last-bag fill from dextrose-based solutions to icodextrin
- d. Increase the number of night-time exchanges to six

Answer: a

Rationale: In the present case it is clear that there is need to increase the 24-hour peritoneal dialysate drain volume in order to increase in peritoneal urea clearances. Options a, c, and d can induce higher 24-hour peritoneal dialysate drain volumes, nevertheless, the addition of a day-exchange will induce the greatest increase in small solute clearances (option A). Therefore, the present PD prescription can be modified to increase significantly small solute clearances and transfer to hemodialysis is not required.